

ON THE TORSION IN THE CHOW MOTIVE OF AN ENRIQUES SURFACE

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ABSTRACT. The integral Chow motive of an Enriques surface S is determined by a torsion summand $\mathfrak{t}(S)$, which contributes the torsion in the cohomology of S . We study isomorphisms between and endomorphisms of motives of the form $\mathfrak{t}(S)$. This is closely related to the integral Hodge conjecture on products of two Enriques surfaces.

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INTRODUCTION

The structure of the category of Chow motives with $\mathbb{Z}$ -coefficients is extremely rich and displays phenomena not present in the more frequently studied $\mathbb{Q}$ -linear category. One such phenomenon, which has been known for a while, is the existence of nontrivial torsion objects, that is, objects M for which $(\mathrm{End}(M), +)$ is a torsion Abelian group. The first examples were discovered by Gorchinskiy–Orlov ([GO13, Proposition 2.3]), who showed that for a smooth projective surface S over $\mathbb{C}$ satisfying $p = q_g = 0$ as well as $\mathrm{CH}_0(S) \cong \mathbb{Z}$ (the conclusion of Bloch’s conjecture ([Blo75])), there is a decomposition of the Chow motive with $\mathbb{Z}$ -coefficients

$$\mathfrak{h}(S) = \mathbf{1} \oplus \mathbf{1}(-1)^{\rho(S)} \oplus \mathbf{1}(-2) \oplus \mathfrak{t}(S)$$

for some torsion Chow motive $\mathfrak{t}(S)$, where $\rho(S)$ is the Picard number of S . Moreover, using Merkurjev–Suslin’s theorem ([MS82]), they were able to compute that the torsion order of $(\mathrm{End}(\mathfrak{t}(S)), +)$ is equal to that of $\mathrm{Pic}(S)_{\mathrm{tors}}$ (see also [Kah17]).

There now arise several basic questions about the structure of these motives $\mathfrak{t}(S)$ and their relationships to each other; for example:

- (a) When is $\mathfrak{t}(S)$ decomposable or indecomposable?
- (b) When are $\mathfrak{t}(S)$ and $\mathfrak{t}(S')$ isomorphic?
- (c) Can we compute the Abelian group $\mathrm{Hom}(\mathfrak{t}(S), \mathfrak{t}(S'))$?

These questions were, to some extent, addressed by Vishik in the paper [Vis23b], which contains a detailed study of torsion motives and numerous interesting results about them. In particular, Vishik showed that p -torsion direct summands of the motives of surfaces are classified, up to isomorphism, by Rost cycle modules of a specific form ([Vis23b, Theorem 4.13]), and used this to obtain information about the endomorphism rings of such motives ([Vis23b, Proposition 4.24 and Remark 4.25]). But it remained unclear how to answer the questions (a)–(c) in any specific concrete setting (aside from trivial cases, such as when $\mathrm{Pic}(S)_{\mathrm{tors}} \not\cong \mathrm{Pic}(S')_{\mathrm{tors}}$). The *raison d'être* of this paper is to remedy this situation and bring further attention to this interesting class of motives.

An “Enriques surface” over a field k is a minimal smooth projective surface S over k of Kodaira dimension 0 and second Betti number $b_2(S) = 10$. These are known to satisfy the hypotheses of [GO13, Proposition 2.3] when $k = \mathbb{C}$, so $\mathfrak{t}(S)$ exists in this case. In fact, as noticed in [SY25, §3], the same argument produces a decomposition $\mathfrak{h}(S)_{\mathbb{Z}[1/p]} = \mathbf{1} \oplus \mathbf{1}(-1)^{10} \oplus \mathbf{1}(-2) \oplus \mathfrak{t}(S)$, where $\mathfrak{h}(S)_{\mathbb{Z}[1/p]}$ denotes the Chow motive of S with $\mathbb{Z}[1/p]$ -coefficients, for any algebraically closed field k of exponential characteristic p (that is, $p := 1$ if $\mathrm{char}(k) = 0$, and otherwise $p := \mathrm{char}(k)$). Enriques surfaces are natural to consider because, from the perspective of Kodaira dimension, they are the simplest surfaces admitting such a decomposition in which $\mathfrak{t}(S)$ is nontrivial.

Summary of main results. We give partial answers to the questions (a)–(c) above; see §1 for detailed statements and an overview of their proofs. Briefly, over any algebraically closed field of characteristic not 2 other than the algebraic closure of a finite field, we

- (a) show that $\mathfrak{t}(S)$ may be decomposable.
- (b) show that $\mathfrak{t}(S) \cong \mathfrak{t}(S')$ for “many” pairs of non-isomorphic Enriques surfaces (S, S') , while $\mathfrak{t}(S) \not\cong \mathfrak{t}(S')$ for “many” pairs (S, S') , even some having the same K3 cover.
- (c) completely determine $\mathrm{Hom}(\mathfrak{t}(S), \mathfrak{t}(S'))$ for “many” (S, S') .

Moreover, nearly all our examples of these phenomena are completely explicit.

We now compare our results with some results in literature. First, the fact that $\mathfrak{t}(S)$ may be decomposable when k is of characteristic 0 directly contradicts [Vis23b, Proposition 4.21]. In Appendix A, we explain what goes wrong in the argument of *loc. cit.* In turn, the statements in [SY25, §7.3] are also wrong (see Remark 6.3 below): for such S , we have $\mathrm{CH}_0(S_{k(S)}) = \mathbb{Z}/2 \oplus \mathbb{Z}/2$ and $H_{\mathrm{ur}}^3(S \times S, \mathbb{Q}_{\ell}/\mathbb{Z}_{\ell}(2)) = 0$ for any prime $\ell \neq \mathrm{char}(k)$ (answering [Kah17, footnote 1]), and if $k = \mathbb{C}$, $S \times S$ satisfies the integral Hodge conjecture in degree 4. In fact, we have been unable to find a single example of an Enriques surface S for which $\mathfrak{t}(S)$ is indecomposable, or equivalently if $k = \mathbb{C}$, for which $S \times S$ fails the integral Hodge conjecture in degree 4. Nevertheless, we believe that $\mathfrak{t}(S)$ should “usually” be indecomposable.

Second, if $k = \mathbb{C}$, that $\mathfrak{t}(S) \not\cong \mathfrak{t}(S')$ implies $S \times S'$ fails the integral Hodge conjecture in degree 4. Such failure was first demonstrated by Diaz in [Dia23], based on arguments of Colliot-Thélène–Voisin ([CTV12]) and Gabber ([CT02, Appendice]; see also [CT19]), and in fact it is not hard to deduce the statement that $\mathfrak{t}(S) \not\cong \mathfrak{t}(S')$ directly from Diaz’s argument. However, one of the aims of this paper is to find examples which are as explicit and numerous as possible, and we succeed in extending Diaz’s argument to apply in more situations. This relies on a computation of the Brauer pullback $\mathrm{Br}(S) \rightarrow \mathrm{Br}(X)$, where $X \rightarrow S$ is the K3 cover, and to this end, we give a complete description of this map (Corollary 2.8) for all S in a certain 2-dimensional family of Enriques surfaces, namely those defined by a Lieberman involution (see (1.1)).

Acknowledgements. We are very grateful to Alexander Vishik for discussing the material of [Vis23b] with us and for providing feedback on an early draft of this paper. We also thank Igor Dolgachev and Daniel Huybrechts for helpful conversations. This material is based upon work supported by the National Science Foundation under Grant No. DMS-2231565.

Statement of AI use. LLMs were used to find typos, check for mathematical accuracy, search for references, and to better understand some of the background material. The initial draft of this paper was generated entirely by the authors.

Notation. Throughout this paper, we will observe the following conventions.

- We denote by k an algebraically closed field and p the exponential characteristic of k .
- If E is an elliptic curve over k and n an integer not divisible by $\mathrm{char}(k)$, we will very often identify the k -group scheme $E[n]$ with its set of k -points.
- All sheaves are étale sheaves, and H^* and π_1 always stand for étale cohomology and the étale fundamental group, respectively. (Of course, for a field K , we will sometimes find it useful to identify $\pi_1(\mathrm{Spec}(K))$ with the absolute Galois group $\mathrm{Gal}(\overline{K}/K)$ and $H^*(\mathrm{Spec}(K))$ with the group cohomology of $\mathrm{Gal}(\overline{K}/K)$).
- For a scheme X , we write $\mathrm{Br}(X) := H^2(X, \mathbb{G}_m)_{\mathrm{tors}}$ for the cohomological Brauer group of X .
- We will, in certain situations, be careful to distinguish between Tate twists, mainly cohomology with $\mathbb{Z}/2$ -coefficients versus μ_2 -coefficients. Since $\mathbb{Z}/2$ and μ_2 are uniquely isomorphic over a field of characteristic not 2, the difference is often immaterial. However, one situation in which it is necessary (to obtain the correct statement of Theorem 1.8) for us to have a precise understanding of the relationship between $\mathbb{Z}/2$ - and μ_2 -cohomology is explained in (2.4).

1. STATEMENTS OF MAIN RESULTS

In this section, k is an algebraically closed field of characteristic not 2. Our main theorems deal with the following class of Enriques surfaces.

1.1. Let $\mathfrak{L}(k)$ denote the set of isomorphism classes of tuples (E, P, F, Q) in which E and F are elliptic curves over k and $P \in E[2] \setminus \{O\}$ and $Q \in F[2] \setminus \{O\}$. To ease notation, if we

write something like “ $\circ \in \mathfrak{L}$ ”, then $(E_\circ, P_\circ, F_\circ, Q_\circ)$ will denote the tuple corresponding to $\circ$. Now fix $\circ \in \mathfrak{L}(k)$. Recall that the Kummer surface $\text{Kum}(E_\circ \times F_\circ)$ is the minimal resolution of the quotient of $(E_\circ \times F_\circ)/\iota$, where ι multiplication by -1 ; alternatively, ι lifts uniquely to an involution of the blowup $\text{Bl}_Z(E_\circ \times F_\circ)$ along the ι -fixed locus $Z := E_\circ[2] \times F_\circ[2]$, and $\text{Kum}(E_\circ \times F_\circ)$ is the quotient $\text{Bl}_Z(E_\circ \times F_\circ)/\iota$. The involution of $E_\circ \times F_\circ$ defined by $(x, y) \mapsto (x + P_\circ, -y + Q_\circ)$ descends to a fixed-point-free involution $\sigma_\circ$ of $\text{Kum}(E_\circ \times F_\circ)$, a “Lieberman involution”, and we denote by $S_\circ := \text{Kum}(E_\circ \times F_\circ)/\sigma_\circ$ the corresponding Enriques surface. We will sometimes say that such an Enriques surface is of “Lieberman type”.

1.2. Theorem. *Let k be an algebraically closed field of characteristic not 2.*

- (a) *Let $\circ, \bullet \in \mathfrak{L}(k)$, and assume there exist isogenies $E_\circ \rightarrow E_\bullet$ and $F_\circ \rightarrow F_\bullet$ of odd degree sending $P_\circ$ to $P_\bullet$ and $Q_\circ$ to $Q_\bullet$, respectively. Then $\mathfrak{t}(S_\circ) \cong \mathfrak{t}(S_\bullet)$.*
- (b) *Let $E_\circ$ be an elliptic curve over k such that $\text{End}(E_\circ) \cong \mathcal{O}_L$ for some quadratic imaginary number field L in which 2 splits. Then there exists $P_\circ \in E_\circ[2] \setminus \{O\}$ such that $\mathfrak{t}(S_\circ)$ is decomposable for any $\circ \in \mathfrak{L}(k)$ of the form $\circ = (E_\circ, P_\circ, F_\circ, Q_\circ)$.*

1.3. Remark. In order for (a) of Theorem 1.2 to be interesting, we need to explain why $S_\circ$ and $S_\bullet$ can be chosen to be not isomorphic as varieties. When $k = \mathbb{C}$, this follows from the general fact that for a K3 surface X , there are only finitely many Abelian varieties A satisfying $X \cong \text{Kum}(A)$ by [HLOY03, Theorem 0.1(1)]. Combining *loc. cit.* with [Orl02, Theorem 2.19] and a short argument, one can show that if $k = \mathbb{C}$, $E_\circ$ and $F_\circ$ are not isogenous, $\text{End}(E_\circ) \cong \mathbb{Z}$,¹ and either $E_\circ \not\cong E_\bullet$ or $E_\circ \not\cong F_\bullet$, then $S_\circ \not\cong S_\bullet$. Over a general algebraically closed field of characteristic not 2, one can use the facts that any Enriques surface admits at most 527 elliptic fibrations up to isomorphism ([BGA24, Theorem 5.4]) and $S_\circ$ admits two elliptic fibrations with respective half-fiber isomorphic to $E_\circ/P_\circ$ or $F_\circ/Q_\circ$.

1.4. Remark. More can be said about the decomposition of 1.2(b); see part (c) of Lemma 6.2. In particular, it yields the first example of a nontrivial Chow motive M over $\mathbb{C}$ with $\mathbb{Z}$ -coefficients having trivial total Chow group, i.e. $\text{CH}^i(M) = 0$ for all i . By contrast, a nontrivial Chow motive over $\mathbb{C}$ with $\mathbb{Q}$ -coefficients has nontrivial total Chow group ([GG12, Lemma 2.1]), and therefore any example of such a motive M must be torsion.

In order to state our second main theorem, we will need the following rather unwieldy notation and terminology.

1.5. Given an elliptic curve E over k and $P \in E[2] \setminus \{O\}$, there exists $\lambda \in k \setminus \{0, 1\}$ for which E is isomorphic to the curve E_λ given by the Weierstrass equation $y^2 = x(x-1)(x-\lambda)$ and P , under this isomorphism, is sent to the point $P_\lambda := (0, 0)$ on E_λ (see [Sil09, Ch. III, Proposition 1.7]). There exists exactly one other value with this property, namely $1/\lambda$. So, for $\lambda, \mu \in k \setminus \{0, 1\}$, we will denote by $S_{\lambda, \mu}$ the Enriques surface attached to $(E_\lambda, P_\lambda, E_\mu, P_\mu) \in \mathfrak{L}(k)$ as in (1.1).

¹or, more generally, $\text{End}(E_\circ)$ is isomorphic to an imaginary quadratic order of odd class number

1.6. Let $\lambda \in k \setminus \{0, 1\}$, and let P be the nonzero element in the image of $E_\lambda[2] \rightarrow (E_\lambda/P_\lambda)[2]$. A direct calculation (using, for example, [Sil09, Ch. III, Example 4.5]) gives $(E_\lambda/P_\lambda, P) \cong (E_{\lambda^*}, P_{\lambda^*})$, where $\lambda^* := (\sqrt{\lambda} + 1)^2/(\sqrt{\lambda} - 1)^2$.

1.7. Definition. Given $\lambda, \mu, \nu \in k \setminus \{0, 1\}$, let us say that (λ, μ, ν) is *suitable*² if there exists a subfield k_0 of k and a discrete valuation v on k_0 with residue characteristic not 2 such that $\lambda, \mu, \nu \in k_0$ and at least one of the following holds:

- (a) $v(\lambda) = v(\lambda - 1) = v(\mu) = v(\mu - 1) = 0$ and $v(\nu) \neq 0$.
- (b) For some indexing $\{\lambda, \mu, \nu\} = \{\lambda_1, \lambda_2, \lambda_3\}$, we have $v(\lambda_1 - 1) = 0$, $v(\lambda_2) \neq 0$, and $v(\lambda_3) \neq 0$.
- (c) For some indexing $\{\lambda, \mu, \nu\} = \{\lambda_1, \lambda_2, \lambda_3\}$, we have $v(\lambda) = v(\lambda - 1) = v(\mu) = v(\mu - 1) = 0$, $v(\lambda_3) \neq 0$, and moreover there is no isogeny $\varphi: E_{\overline{\lambda_1}} \rightarrow E_{\overline{\lambda_2}}$ such that $\varphi(P_{\overline{\lambda_1}}) = 0$ and $P_{\overline{\lambda_2}} \in \varphi(E_{\overline{\lambda_1}}[2])$, where the bar denotes the image in a fixed algebraic closure of the residue field of v .

Note that (λ, μ, ν) is *never* suitable if k is the algebraic closure of a finite field.

1.8. Theorem. Let k be an algebraically closed field of characteristic not 2, and fix $\lambda, \mu, \lambda', \mu' \in k \setminus \{0, 1\}$. Assume that one of $(\lambda, \mu, \lambda'^*)$, (λ, μ, μ'^*) , $(\lambda', \mu', \lambda^*)$, or (λ', μ', μ^*) is suitable. Then $\mathfrak{t}(S_{\lambda, \mu}) \not\cong \mathfrak{t}(S_{\lambda', \mu'})$.

1.9. Remark. It is easy to use Theorem 1.8 to find Enriques surfaces S and S' having the same K3 cover but satisfying $\mathfrak{t}(S) \not\cong \mathfrak{t}(S')$; see Example 6.8.

1.10. Remark. In several cases, we are able to exactly compute $\text{Hom}(\mathfrak{t}(S), \mathfrak{t}(S'))$ (Examples 6.5 and 6.6). In fact, we show that four out of the five possibilities of this group (Lemma 6.2) do occur; the missing possibility occurs if and only if some $\mathfrak{t}(S)$ is indecomposable.

1.11. On the proofs. After some preliminaries in §§2–5, the proofs of Theorems 1.2 and 1.8 are completed in §6. The proofs are quite different in nature: The former requires us to produce algebraic cycles, while the latter requires us to show that a certain cohomology class is not algebraic.

The proof of Theorem 1.2 rests on a finiteness result, in the spirit of Merkurjev–Suslin ([MS82, §18]), for torsion in CH^2 (Proposition 4.2). This result puts strong restrictions on morphisms and endomorphisms of torsion summands of the motives of smooth projective surfaces ((5.2)–(5.3)). In particular, we observe that Theorem 1.2 is implied by the algebraicity of certain 2-torsion cohomology classes in $H^4(S_\circ \times S_\bullet, \mathbb{Z}_2(2))$ (Lemma 6.2). A simple geometric argument shows that rational maps of odd or even degree provide the necessary algebraic cycles (Proposition 6.4). Finally, we construct the rational maps in §6.

The idea of the proof of Theorem 1.8 is to study $\text{Hom}(\mathfrak{t}(S), \mathfrak{h}(E))$ for Enriques surfaces S and elliptic curves E . More precisely, since $H^1(\mathfrak{t}(S), \mu_2) \cong \mathbb{Z}/2$, we get a map $\text{Hom}(\mathfrak{t}(S), \mathfrak{h}(E)) \rightarrow H^1(E, \mu_2)$, and the image of this map is an invariant of $\mathfrak{t}(S)$. Assuming, for example, that $(\lambda, \mu, \lambda^*)$ is suitable, this reduces the claim to the non-algebraicity of a certain 2-torsion class

²by which we mean “suitable for the proof of Corollary 3.4”

in $H^4(S_{\lambda,\mu} \times E_{\lambda'^*}, \mathbb{Z}_2(2))$. By the work of Colliot-Thélène–Voisin ([CTV12, Théorème 3.7]; see Proposition 4.3), it suffices to show that a certain class in $H^3(S_{\lambda,\mu} \times E_{\lambda'^*}, \mu_2^{\otimes 2})$ is nonvanishing in Galois cohomology. For this, we use modified version of a degeneration argument of Gabber ([CT02, Appendice], [AS24, Theorem 3.3]; see Proposition 3.2), supplemented by the explicit calculation of the Brauer pullback for Enriques surfaces of Lieberman type (Proposition 2.5). We remark that the integral Hodge conjecture for varieties of the form $S \times E$ was first shown to be false by Benoist–Ottem in [BO20, Theorem 0.1] using a Hilbert-scheme argument.

1.12. Remark. Aside from finding an example when $\mathfrak{t}(S)$ is indecomposable, the other major shortcoming of our results is that we are unable to find a single pair of Enriques surfaces S and S' over the algebraic closure of a finite field such that $\mathfrak{t}(S) \not\cong \mathfrak{t}(S')$, because the argument of Proposition 3.2 relies on passing to the residue field of a (nontrivial) valuation on the base field. What is more, Schoen proved that over such a field, the integral Tate conjecture for 1-cycles is a consequence of the usual Tate conjecture for divisors on surfaces ([Sch98]). Thus we expect that the map $\mathrm{Hom}(\mathfrak{t}(S), \mathfrak{h}(E)) \rightarrow H^1(E, \mu_2)$ is surjective for all S and E , rendering the motivic invariants $\mathrm{Img}(\mathrm{Hom}(\mathfrak{t}(S), \mathfrak{h}(E)) \rightarrow H^1(E, \mu_2))$ completely useless.

2. BRAUER CLASSES ON ENRIQUES SURFACES OF LIEBERMAN TYPE

In this section, k is an algebraically closed field of characteristic not 2. If S is an Enriques surface over k , then $\mathrm{Br}(S) \cong \mathbb{Z}/2$ ([CDL25, Theorem 1.2.17]). In [Bea09], Beauville gives an explicit lattice-theoretic description of the locus of complex Enriques surfaces S for which the map $\mathrm{Br}(S) \rightarrow \mathrm{Br}(X)$ vanishes, where $X \rightarrow S$ is the K3 cover. In subsequent work, such as [GS12], explicit families of Enriques surfaces with vanishing Brauer pullback were exhibited. The purpose of this section is to compute $\mathrm{Br}(S) \rightarrow \mathrm{Br}(X)$ for any Enriques surface S of Lieberman type over k . Note that the Enriques surfaces in [GS12] are also constructed using products of elliptic curves; we have not tried to determine the relationship between them and ones of Lieberman type.

2.1. Let S be an Enriques surface over k . Then, from the μ_2 -Kummer sequence and the fact that $\mathrm{Br}(S) \cong \mathbb{Z}/2$, we get a surjective map $H^2(S, \mu_2) \twoheadrightarrow \mathrm{Br}(S)$ whose kernel is the group of algebraic classes. A comparison of the μ_2 - and μ_4 -Kummer sequences shows that a class in $H^2(S, \mu_2)$ is algebraic if and only if it lifts to a class in $H^2(S, \mu_4)$.

2.2. First, a reminder on the cohomology of a blowup. Let $\iota: Z \hookrightarrow X$ be a codimension- c closed immersion of smooth k -varieties, let $\pi: \mathrm{Bl}_Z(X) \rightarrow X$ denote the blowup, and let $\iota': Z' \hookrightarrow \mathrm{Bl}_Z(X)$ denote the inclusion of the exceptional divisor. Then the map

$$\begin{pmatrix} \pi^* & \iota'_* \\ 0 & (\pi|_{Z'})_* \end{pmatrix} : H^i(X, \mu_2^{\otimes j}) \oplus H^{i-2}(Z', \mu_2^{\otimes j-1}) \rightarrow H^i(\mathrm{Bl}_Z(X), \mu_2^{\otimes j}) \oplus H^{i-2c}(Z, \mu_2^{\otimes j-c})$$

is an isomorphism ([sga77, Exposé VI, Corollaire 8.3]). This is the “blowup formula”.

2.3. We will use the notation and terminology of (1.1). Fix $(E, P, F, Q) \in \mathfrak{L}(k)$. Set $Z := E[2] \times F[2]$ and $Y^\circ := (E \times F) \setminus Z$, and let Y denote the blowup of $E \times F$ along Z . Set

$X := \text{Kum}(E \times F)$, and let $X \rightarrow S$ be the quotient by the Lieberman involution attached to (E, P, F, Q) . We get a commutative diagram

$$\begin{array}{ccccccc} H^2(S, \mu_2) & \rightarrow & H^2(X, \mu_2) & \rightarrow & H^2(Y, \mu_2) & \rightarrow & H^2(Y^\circ, \mu_2) \xleftarrow{\sim} H^2(E \times F, \mu_2) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \text{Br}(S) & \longrightarrow & \text{Br}(X) & \longrightarrow & \text{Br}(Y) & \xleftarrow{\sim} & \text{Br}(E \times F); \end{array}$$

the last arrow on the top is an isomorphism by the Gysin sequence ([Mil80, Ch. VI, Remark 5.4(b)]) since $\text{codim}_{E \times F}(Z) = 2$, and the last arrow on the bottom can be checked to be an isomorphism using the Kummer sequence and the blowup formula (2.2).

2.4. Let E be an elliptic curve over k . Recall that $H^1(E, \mathbb{Z}/2)$ is isomorphic to the group of étale $\mathbb{Z}/2$ -torsors over E .

- (a) We will frequently make use of the isomorphism $\text{Hom}(E[2], \mathbb{Z}/2) \xrightarrow{\sim} H^1(E, \mathbb{Z}/2)$, which we will denote $\theta \mapsto \alpha_\theta$, characterized as follows: For a surjection $\theta: E[2] \twoheadrightarrow \mathbb{Z}/2$, α_θ is given by the isogeny $E/\text{Ker}(\theta) \rightarrow E$ dual to the quotient isogeny. In turn, using the perfect Poincaré pairing $H^1(E, \mathbb{Z}/2) \times H^1(E, \mu_2) \rightarrow \mathbb{Z}/2$, we get an isomorphism $E[2] \xrightarrow{\sim} H^1(E, \mu_2)$, which we will also denote by $P \mapsto \alpha_P$.
- (b) Given $P \in E[2] \setminus \{O\}$, there is a unique homomorphism $P^\vee: E[2] \xrightarrow{\sim} \mathbb{Z}/2$ with kernel $\{O, P\}$. Using that the pairing on $H^1(E, \mu_2)$ is alternating, one checks that the isomorphism $H^1(E, \mathbb{Z}/2) \xrightarrow{\sim} H^1(E, \mu_2)$ induced by $\mathbb{Z}/2 \xrightarrow{\sim} \mu_2$ sends $\alpha_{P^\vee}$ to α_P for each $P \in E[2] \setminus \{O\}$.

2.5. Proposition. *In the setting of (2.3), let $\beta \in H^2(S, \mu_2)$ be any non-algebraic class. Then the image of β under*

$$H^2(S, \mu_2) \rightarrow H^2(E \times F, \mu_2) \rightarrow H^1(E, \mathbb{Z}/2) \otimes H^1(F, \mu_2)$$

equals $\alpha_{P^\vee} \otimes \alpha_Q$ in the notation of 2.4(a).

Before giving the proof of Proposition 2.5, we provide some reminders on étale local systems.

2.6. Let T be a scheme. An étale sheaf on T is called *lcc* if it is locally constant and has finite stalks. For such a sheaf $\mathcal{F}$, the stalk $\mathcal{F}_{\bar{t}}$ at a geometric point $\bar{t}$ has a “monodromy” action of $\pi_1(T, \bar{t})$ which is functorial in $(T, \bar{t})$. If T is connected, $\mathcal{F} \mapsto \mathcal{F}_{\bar{t}}$ defines an equivalence of categories ([sta26, Section ODV4])

$$(\text{lcc sheaves of sets on } T) \xrightarrow{\sim} (\text{continuous actions of } \pi_1(T, \bar{t}) \text{ on finite sets}).$$

Under this equivalence, constant sheaves correspond to π_1 -sets with the trivial action. More generally, the image of the (injective) map $\Gamma(T, \mathcal{F}) \rightarrow \mathcal{F}_{\bar{t}}$ is the set of π_1 -invariants. If $T = T_1 \times_{\Sigma} T_2$ and $\mathcal{F}_i$ is an lcc Abelian sheaf on T_i , we denote by $\mathcal{F}_1 \boxtimes \mathcal{F}_2$ the “external product” sheaf $\text{pr}_1^{-1} \mathcal{F}_1 \otimes \text{pr}_2^{-1} \mathcal{F}_2$ on T , which is again lcc. If $\bar{t} = (\bar{t}_1, \bar{t}_2)$, then $(\mathcal{F}_1 \boxtimes \mathcal{F}_2)_{\bar{t}} \cong \mathcal{F}_{1, \bar{t}_1} \otimes \mathcal{F}_{2, \bar{t}_2}$.

2.7. Let T be a scheme. For a smooth proper morphism $f: \mathcal{X} \rightarrow T$ and an lcc Abelian sheaf $\mathcal{F}$ on $\mathcal{X}$ whose stalks have order invertible on T , we will write $\mathcal{H}^i(\mathcal{X}/T, \mathcal{F}) := R^i f_* \mathcal{F}$. By the theorems on smooth and proper base change ([Mil80, Ch. VI, Corollary 4.2]), this is an lcc

Abelian sheaf on T , and at a geometric point $\bar{t}$, its stalk $\mathcal{H}^i(\mathcal{X}/T, \mathcal{F})_{\bar{t}}$ is naturally isomorphic to $H^i(\mathcal{X}_{\bar{t}}, \iota^{-1}\mathcal{F})$, where ι denotes the inclusion $\mathcal{X}_{\bar{t}} \hookrightarrow \mathcal{X}$. If $T = T_1 \times_{\Sigma} T_2$, $\mathcal{X} = \mathcal{X}_1 \times_{\Sigma} \mathcal{X}_2$ for smooth proper T_i -schemes $\mathcal{X}_i$, and $\mathcal{F} = \mathcal{F}_1 \boxtimes \mathcal{F}_2$ for lcc Abelian sheaves $\mathcal{F}_i$ on $\mathcal{X}_i$ of order invertible on T_i , the Künneth map

$$\bigoplus_{a+b=i} \mathcal{H}^a(\mathcal{X}_1/T_1, \mathcal{F}_1) \boxtimes \mathcal{H}^b(\mathcal{X}_2/T_2, \mathcal{F}_2) \rightarrow \mathcal{H}^i(\mathcal{X}/T, \mathcal{F})$$

is an isomorphism of sheaves, because it is an isomorphism on stalks by the usual Künneth theorem ([Mil80, Ch. VI, Corollary 8.13]).

Proof of Proposition 2.5. Let $\pi: Y \rightarrow E \times F$ denote the blowup morphism. We first claim that it is equivalent to compute the image of β under

$$H^2(S, \mu_2) \rightarrow H^2(Y, \mu_2) \xrightarrow{\pi_*} H^2(E \times F, \mu_2) \rightarrow H^1(E, \mathbb{Z}/2) \otimes H^1(F, \mu_2).$$

For this, we will show that the triangle

$$\begin{array}{ccc} H^2(Y, \mu_2) & \xrightarrow{\pi_*} & H^2(E \times F, \mu_2) \\ & \searrow u'^* & \swarrow u^* \\ & H^2(Y^\circ, \mu_2) & \end{array}$$

commutes, where u and u' are the respective open embeddings. Let ι be the inclusion $Z \hookrightarrow E \times F$, and let $\iota': Z' \hookrightarrow Y$ be the inclusion of the exceptional divisor. Then we have $u'^* \circ \pi^* = u^*$ and $\pi_* \circ \iota'_* = \iota_* \circ (\pi|_{Z'})_*$ by functoriality, $u^* \circ \iota_* = 0$ and $u'^* \circ \iota'_* = 0$ by the Gysin sequence ([Mil80, Ch. VI, Remark 5.4(b)]; we also are using [Mil80, Ch. VI, Remark 11.6(b)]), and $\pi_* \circ \pi^* = \text{id}$ because π is birational. Now fix $\xi \in H^2(Y, \mu_2)$. By the blowup formula (2.2), we can (uniquely) write $\xi = \pi^*\xi_0 + \iota'_*\zeta$ for some $\xi_0 \in H^2(E \times F, \mu_2)$ and $\zeta \in H^0(Z', \mu_2)$. Then

$$u'^*\xi = u'^*\pi^*\xi_0 + u'^*\iota'_*\zeta = u^*\xi_0$$

and

$$u^*\pi_*\xi = u^*\pi_*\pi^*\xi_0 + u^*\pi_*\iota'_*\zeta = u^*\xi_0 + u^*\iota_*(\pi|_{Z'})_*\zeta = u^*\xi_0,$$

proving the commutativity of the triangle.

The open modular curve $Y_1(4)$, the fine moduli space of elliptic curves with a marked point of order 4, exists as a connected scheme over $\mathbb{Z}[1/2]$ ([Mum65]);³ let $\mathcal{E} \rightarrow Y_1(4)$ denote its universal elliptic curve. Set $T := Y_1(4) \times_{\mathbb{Z}} Y_1(4)$, and let $\mathcal{Y} \rightarrow \mathcal{E} \times_{\mathbb{Z}} \mathcal{E}$ be the blowup along the 2-torsion locus $\mathcal{Z} := \mathcal{E}[2] \times_{\mathbb{Z}} \mathcal{E}[2]$. Since both $\mathcal{E} \times_{\mathbb{Z}} \mathcal{E}$ and $\mathcal{Z}$ are smooth and proper over T , so is $\mathcal{Y}$, and for any $T' \rightarrow T$, the base-change $\mathcal{Y}_{T'}$ is naturally identified with the blowup of $(\mathcal{E} \times_{\mathbb{Z}} \mathcal{E})_T$ along $\mathcal{Z}_T$. Finally, we can define quotients $\mathcal{Y} \rightarrow \mathcal{X} \rightarrow \mathcal{S}$ by the Kummer and Lieberman involutions, respectively, where for the latter we use that $\mathcal{E}$ has a distinguished 2-torsion section, namely twice the distinguished 4-torsion section. To summarize, $\mathcal{Y} \rightarrow \mathcal{S}$ is the morphism of smooth proper T -schemes whose fiber over $((E, P'), (F, Q')) \in T(k)$, where $2P' = P$ and $2Q' = Q$, is the map $Y \rightarrow S$ defined in (2.3).

³It would be more natural to argue with $Y_1(2)$ instead, but this would require viewing $Y_1(2)$ as a stack; we prefer to use only schemes in this paper.

By the first paragraph, we are interested in the composition

$$(2.7.1) \quad \begin{aligned} \mathcal{H}^2(\mathcal{S}/T, \mu_2) &\rightarrow \mathcal{H}^2(\mathcal{Y}/T, \mu_2) \xrightarrow{\pi_*} \mathcal{H}^2(\mathcal{E} \times_{\mathbb{Z}} \mathcal{E}/T, \mu_2) \\ &\rightarrow \mathcal{H}^1(\mathcal{E}/Y_1(4), \mathbb{Z}/2) \boxtimes \mathcal{H}^1(\mathcal{E}/Y_1(4), \mu_2), \end{aligned}$$

where π now denotes the T -morphism $\mathcal{Y} \rightarrow \mathcal{E} \times_{\mathbb{Z}} \mathcal{E}$. Notice that $\mathcal{H}^2(\mathcal{S}/T, \mu_2)$ admits an lcc subsheaf $\mathcal{G}$ whose stalk at a geometric point $\bar{t}$ of T is the space of algebraic classes in $H^2(\mathcal{S}_{\bar{t}}, \mu_2)$; indeed, the image of $\mathcal{H}^2(\mathcal{S}/T, \mu_4) \rightarrow \mathcal{H}^2(\mathcal{S}/T, \mu_2)$ has this property, as noted in (2.1). We claim that $\mathcal{G}$ is included in the kernel of (2.7.1). Indeed, when $\bar{t} = (\bar{t}_1, \bar{t}_2)$ and the corresponding elliptic curves $\mathcal{E}_{\bar{t}_1}$ and $\mathcal{E}_{\bar{t}_2}$ are not isogenous, the only algebraic classes in $H^2(\mathcal{E}_{\bar{t}_1} \times \mathcal{E}_{\bar{t}_2}, \mu_2)$ are classes in the $(0, 2)$ - and $(2, 0)$ -pieces of the Künneth decomposition. Since every sheaf in sight is locally constant and T is connected, inclusion at one stalk implies inclusion everywhere, proving the claim. Therefore, (2.7.1) induces a map

$$(2.7.2) \quad \mathcal{H}^2(\mathcal{S}/T, \mu_2)/\mathcal{G} \rightarrow \mathcal{H} := \mathcal{H}^1(\mathcal{E}/Y_1(4), \mathbb{Z}/2) \boxtimes \mathcal{H}^1(\mathcal{E}/Y_1(4), \mu_2).$$

But each stalk of $\mathcal{H}^2(\mathcal{S}/T, \mu_2)/\mathcal{G}$ is isomorphic to $\text{Br}(S)[2] \cong \mathbb{Z}/2$, so this sheaf is constant (because $\mathbb{Z}/2$ has no nontrivial π_1 -action), whence the image of (2.7.2) defines a global section of $\mathcal{H}$. Thus we will be done if we can verify the following:

- (a) The map (2.7.2) is nonzero.
- (b) The sheaf $\mathcal{H}$ has just one nonzero global section, and the stalk of this section over $((E, P'), (F, Q')) \in T(k)$ is the one described in the statement of the present proposition.

To prove (a), it suffices to check that $\text{Br}(\mathcal{S}_{\bar{t}}) \rightarrow \text{Br}(\mathcal{Y}_{\bar{t}})$ is nonzero for some $\bar{t}$. As a consequence of [SV23, Corollary B(i)], there exists $\bar{t} \in T(\mathbb{C})$ for which the pullback map $\text{Br}(\mathcal{S}_{\bar{t}}) \rightarrow \text{Br}(\mathcal{X}_{\bar{t}})$ is nonzero, where $\mathcal{X}_{\bar{t}} \rightarrow \mathcal{S}_{\bar{t}}$ is the K3 cover. Examining the Hochschild–Serre spectral sequence $E_2^{a,b} := H^a(\mathbb{Z}/2, H^b(\mathcal{Y}_{\bar{t}}^{\circ}, \mathbb{G}_m)) \Rightarrow H^{a+b}(\mathcal{X}_{\bar{t}}^{\circ}, \mathbb{G}_m)$, one sees that the pullback map $\text{Br}(\mathcal{X}_{\bar{t}}) \rightarrow \text{Br}(\mathcal{Y}_{\bar{t}})$ is an isomorphism (see [SZ12, Proposition 1.3]). Thus (a) holds.

Now we prove (b). It is clear that $\mathcal{H}$ has a global section with the stated description. There are several ways to see that it is the only nonzero global section; here is one of them. Consider the elliptic curves E_0 and F_0 over $\mathbb{Q}$ given, respectively, by the equations

$$y^2 = x(x^2 + 4) \quad \text{and} \quad y^2 = x(x^2 - x + 1).$$

We have $E_0[4](\mathbb{Q}) \cong F_0[4](\mathbb{Q}) \cong \mathbb{Z}/4$ (see [lmf26, elliptic curves 32.a4 and 24.a5]). In addition, the action of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ on $E_0[2]$ (by which we mean $E_0[2](\overline{\mathbb{Q}})$) visibly factors through $\text{Gal}(\mathbb{Q}(\sqrt{-1})/\mathbb{Q})$, the nontrivial element of this group acting via the matrix

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

(with respect to a suitable basis); the same is true of $F_0[2]$, except that the action factors through $\text{Gal}(\mathbb{Q}(\sqrt{-3})/\mathbb{Q})$. Therefore, the action of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ on $\text{Hom}(E_0[2], F_0[2])$ factors through $\text{Gal}(\mathbb{Q}(\sqrt{-1}, \sqrt{-3})/\mathbb{Q}) \cong \text{Gal}(\mathbb{Q}(\sqrt{-1})/\mathbb{Q}) \times \text{Gal}(\mathbb{Q}(\sqrt{-3})/\mathbb{Q})$, and it is easily seen

that there is just one nonzero invariant element (namely, the map with kernel $\{0, P\}$ and image $\{0, Q\}$, where P and Q are the nonzero elements of $E_0[2](\mathbb{Q})$ and $F_0[2](\mathbb{Q})$, respectively).

Let $t \in T(\mathbb{Q})$ be the point determined by E_0 and F_0 , and let $\bar{t} \in T(\overline{\mathbb{Q}})$ be a geometric point above t . We get a homomorphism $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \cong \pi_1(\text{Spec}(\mathbb{Q}), \text{Spec}(\overline{\mathbb{Q}})) \rightarrow \pi_1(T, \bar{t})$ under which the monodromy action of $\pi_1(T, \bar{t})$ on $\mathcal{H}_t \cong H^1(E_{0, \overline{\mathbb{Q}}}, \mathbb{Z}/2) \otimes H^1(F_{0, \overline{\mathbb{Q}}}, \mu_2)$ pulls back to the tensor product of the usual Galois actions on étale cohomology. We saw in the previous paragraph that this action has one nonzero invariant, hence $\mathcal{H}$ has at most one nonzero global section, as desired. $\blacksquare$

2.8. Corollary. *In the setting of (2.3), the maps $\text{Br}(S) \rightarrow \text{Br}(X)$ and $\text{Br}(S) \rightarrow \text{Br}(E \times F)$ vanish if and only if there exists an isogeny $\varphi: E \rightarrow F$ with $\varphi(P) = 0$ and $Q \in \varphi(E[2])$.*

Proof. By [SZ12, Proposition 1.3], $\text{Br}(S) \rightarrow \text{Br}(X)$ vanishes if and only if $\text{Br}(S) \rightarrow \text{Br}(E \times F)$ vanishes. Since $\text{Br}(S) \cong \mathbb{Z}/2$, the map $\text{Br}(S) \rightarrow \text{Br}(E \times F)$ vanishes if and only if, in the notation of Proposition 2.5, the image of β in $H^2(E \times F, \mu_2)$ is algebraic. Because of the identification $H^1(E, \mathbb{Z}/2) \otimes H^1(F, \mu_2) \cong \text{Hom}(E[2], F[2])$ of 2.4(a), the image of β is algebraic if and only if such an isogeny φ exists. $\blacksquare$

3. NONVANISHING OF SOME CLASSES IN GALOIS COHOMOLOGY

The goal of this section is to give criteria (Corollary 3.4) for certain classes in $H^3(S \times E, \mu_2)$ to be nonvanishing in Galois cohomology, where S is an Enriques surface defined by a Lieberman involution and E an elliptic curve. The first such criterion was given by Diaz in [Dia23, §4]; we find two more criteria by using Proposition 2.5 above. The key input, Proposition 3.2, is modeled after [AS24, Theorem 3.3], which is in turn a modified version of an argument of Gabber ([CT02, Appendice]).

3.1. Let (K, v) be a discretely valued field with residue field κ , let n be a positive integer not divisible by $\text{char}(\kappa)$, and let A be an unramified n -torsion finite Gal_K -module. Then there exists a “residue map” in Galois cohomology $r_v: H^i(K, A \otimes \mu_n) \rightarrow H^{i-1}(\kappa, A)$. It can be written down explicitly at the level of cochains in group cohomology ([Ser97, II, Appendix, §1]) or defined by the Gysin sequence in étale cohomology ([CT95, §3.3]). The following can be checked at the level of cochains in group cohomology.

- (a) The residue map is natural in (K, v) in the sense that if K'/K is a field extension and v' is a discrete valuation on K' such that $v'|_K = v$, then the square

$$\begin{array}{ccc} H^i(K, A \otimes \mu_n) & \xrightarrow{r_v} & H^{i-1}(\kappa, A) \\ \downarrow & & \downarrow \\ H^i(K', A \otimes \mu_n) & \xrightarrow{r_{v'}} & H^{i-1}(\kappa', A) \end{array}$$

commutes, where κ' denotes the residue field of v' .

- (b) Let $\alpha \in H^1(K, \mu_n)$ and $\xi \in H^i(K, A)$. If the restriction of ξ to the decomposition subgroup of v is pulled back from some $\tilde{\xi} \in H^i(\kappa, A)$, then $r_v(\alpha \smile \xi) = r_v(\alpha) \cdot \tilde{\xi}$.

In the following, given a discrete valuation ring $\mathfrak{o}$, an integral $\mathfrak{o}$ -scheme $\mathcal{X}$ with integral special fiber, and a map $\mathfrak{o} \rightarrow L$ for some field L , we write $L(\mathcal{X})$ for the function field of the base-change $\mathcal{X}_L$.

3.2. Proposition. *Let $X, Y_1, \dots, Y_b$ be integral k -schemes, and let $\xi \in H^a(X, \mu_n^{\otimes a'})$ and $\alpha_i \in H^1(Y_i, \mu_n)$ for each i . Assume that there exist*

- *a discrete valuation subring $\mathfrak{o}$ of k with residue field κ such that $\text{char}(\kappa) \nmid n$,*
- *integral $\mathfrak{o}$ -schemes $\mathcal{X}, \mathcal{Y}_1, \dots, \mathcal{Y}_b$ which recover $X, Y_1, \dots, Y_b$ upon base-change to k and have integral special fibers $\mathcal{X}_\kappa, \mathcal{Y}_{1,\kappa}, \dots, \mathcal{Y}_{b,\kappa}$, and*
- *classes $\xi_\mathfrak{o} \in H^a(\mathcal{X}, \mu_n^{\otimes a'})$ and $\alpha_{i,\mathfrak{o}} \in H^1(\mathcal{Y}_i, \mu_n)$*

such that

- (i) $\xi_\mathfrak{o}, \alpha_{1,\mathfrak{o}}, \dots, \alpha_{b,\mathfrak{o}}$ *recover $\xi, \alpha_1, \dots, \alpha_b$ upon pullback to $X, Y_1, \dots, Y_b$, respectively.*
- (ii) $\xi_\mathfrak{o}$ *does not vanish in $H^a(\kappa'(\mathcal{X}), \mu_n^{\otimes a'})$ for any field extension κ'/κ .*
- (iii) *for each i , the image of $\alpha_{i,\mathfrak{o}}$ under*

$$H^1(\mathcal{Y}_i, \mu_n) \rightarrow H^1(\kappa(\mathcal{Y}_i), \mu_n) \xrightarrow{r_{v_i}} \mathbb{Z}/n$$

is invertible for some valuation v_i on $\kappa(\mathcal{Y}_i)$ induced by a prime divisor of $\mathcal{Y}_{i,\kappa}$.

Then $\xi \boxtimes \alpha_1 \boxtimes \dots \boxtimes \alpha_b$ does not vanish in $H^{a+b}(k(X \times \prod_i Y_i), \mu_n^{\otimes a'+b})$.

Proof. Let $k_0 \subseteq k$ denote the fraction field of $\mathfrak{o}$. We first reduce to the case when k is algebraic over k_0 . Let $\mathfrak{B}$ be a transcendence basis of k/k_0 . Then v extends to a discrete valuation on $k_0(\mathfrak{B})$ by putting

$$v\left(\sum_z a_z \cdot z\right) := \min_z v(a_z),$$

where z ranges over all monomials in the elements of $\mathfrak{B}$ and each a_z lives in k_0 . All of the hypotheses remain true if we replace k_0 by $k_0(\mathfrak{B})$.

Put $\mathcal{Z} := \mathcal{X} \times \prod_i \mathcal{Y}_i$ (product over $\mathfrak{o}$) and $\theta_\mathfrak{o} := \xi_\mathfrak{o} \boxtimes \alpha_{1,\mathfrak{o}} \boxtimes \dots \boxtimes \alpha_{b,\mathfrak{o}} \in H^{a+b}(\mathcal{Z}, \mu_n^{\otimes a'+b})$. We will show that the image of $\theta_\mathfrak{o}$ in $H^{a+b}(k_0(\mathcal{Z}), \mu_n^{\otimes a'+b})$ is nonzero. This suffices to prove the present proposition, because the same argument will show that the image of $\theta_\mathfrak{o}$ remains nonzero when k_0 is replaced by any finite extension, hence also (thanks to the previous paragraph) by k itself.

The prime divisor $\mathcal{Z}_\kappa$ of $\mathcal{Z}$ gives rise to an extension of v to a discrete valuation on $k_0(\mathcal{Z})$ with residue field $\kappa(\mathcal{Z})$. Let $\mathfrak{o}(\mathcal{Z})$ denote its valuation ring, and let $\mathfrak{o}(\mathcal{Z})_v$ and $k_0(\mathcal{Z})_v$ be the respective completions at v . We get a commutative diagram

$$\begin{array}{ccccc} H^{a+b}(\mathcal{Z}_{k_0}, \mu_n^{\otimes a'+b}) & \longrightarrow & H^{a+b}(k_0(\mathcal{Z}), \mu_n^{\otimes a'+b}) & \longrightarrow & H^{a+b}(k_0(\mathcal{Z})_v, \mu_n^{\otimes a'+b}) \\ \uparrow & & \uparrow & & \uparrow (*) \\ \theta_\mathfrak{o} \in H^{a+b}(\mathcal{Z}, \mu_n^{\otimes a'+b}) & \longrightarrow & H^{a+b}(\mathfrak{o}(\mathcal{Z}), \mu_n^{\otimes a'+b}) & \longrightarrow & H^{a+b}(\mathfrak{o}(\mathcal{Z})_v, \mu_n^{\otimes a'+b}) \\ \downarrow & & \downarrow & & \downarrow \sim \\ H^{a+b}(\mathcal{Z}_\kappa, \mu_n^{\otimes a'+b}) & \longrightarrow & H^{a+b}(\kappa(\mathcal{Z}), \mu_n^{\otimes a'+b}) & = & H^{a+b}(\kappa(\mathcal{Z}), \mu_n^{\otimes a'+b}), \end{array}$$

where the arrow $(*)$ is injective by [CT02, Appendice, Lemme A5].⁴ In view of the diagram, it suffices to show that the image θ' of $\theta_{\mathfrak{o}}$ in $H^{a+b}(\kappa(\mathcal{Z}), \mu_n^{\otimes a'+b})$ is nontrivial. Let ξ' and α'_i be, respectively, the images of $\xi_{\mathfrak{o}}$ and $\alpha_{i,\mathfrak{o}}$ in $H^a(\kappa(\mathcal{X}), \mu_n^{\otimes a'})$ and $H^1(\kappa(\mathcal{Y}_i), \mu_n)$, so that $\theta' = \xi' \boxtimes \alpha'_1 \boxtimes \cdots \boxtimes \alpha'_b$. Let W_i be a prime divisor of $\mathcal{Y}_{i,\kappa}$ inducing the valuation v_i of item (iii) of the statement. Then the prime divisor $\mathcal{X}_{\kappa} \times \mathcal{Y}_{1,\kappa} \times \cdots \times \mathcal{Y}_{b-1,\kappa} \times W_b$ of $\mathcal{Z}_{\kappa}$ induces an extension of v_b to a valuation $\tilde{v}_b$ on $\kappa(\mathcal{Z})$, and

$$r_{\tilde{v}_b}(\theta') = r_{v_b}(\alpha'_b) \cdot f^*(\xi' \boxtimes \alpha'_1 \boxtimes \cdots \boxtimes \alpha'_{b-1});$$

here $r_{v_b}(\alpha'_b) \in \mathbb{Z}/n$ is invertible by (iii), and f^* denotes pullback in Galois cohomology from $\kappa(\mathcal{X} \times \mathcal{Y}_1 \times \cdots \times \mathcal{Y}_{b-1})$ to $\kappa(\mathcal{X} \times \mathcal{Y}_1 \times \cdots \times \mathcal{Y}_{b-1} \times W_b)$. Repeatedly taking residues in a similar way, we find that $(r_{\tilde{v}_1} \circ \cdots \circ r_{\tilde{v}_b})(\theta')$ is, up to an element of $(\mathbb{Z}/n)^{\times}$, the pullback of ξ' from $\kappa(\mathcal{X})$ to $\kappa(\mathcal{X} \times W_1 \times \cdots \times W_b) = \kappa(W_1 \times \cdots \times W_b)(\mathcal{X})$, which is nonzero by (ii). ■

3.3. Corollary. *Let k be an algebraically closed field of characteristic not 2. Let E be an elliptic curve over k , S an Enriques surface over k , $\beta \in H^2(S, \mu_2)$ any non-algebraic class, and $P \in E[2] \setminus \{O\}$. Assume that there exists a subfield k_0 of k and a discrete valuation v on k_0 with residue characteristic not 2 such that (E, P) and S descend to k_0 and have, respectively, multiplicative and good reduction mod v , and further that P reduces to a smooth point. Then the image of $\beta \otimes \alpha_P$ in $H^3(k(S \times E), \mu_2^{\otimes 2})$ is nonzero, where α_P is as 2.4(a).*

Proof. This is a slight generalization of the situation of [Dia23, §4], and our argument resembles that of Diaz. We will show that the assumptions of Proposition 3.2 hold with $b = 1$ and $(X, Y_1, \xi, \alpha_1) = (S, E, \beta, \alpha_P)$.

We first check condition (iii). After finitely enlarging k_0 , we will assume that $(E, P) = (E_{\nu}, P_{\nu})$ for some $\nu \in k \setminus \{0, 1\}$. Since E_{ν} has bad reduction, $v(\nu) \neq 0$, and we will assume that $v(\nu) > 0$ after replacing ν with ν^{-1} if necessary. As noted in (1.6) and 2.4(b), the class $\alpha_{P_{\nu}}$ corresponds to the $\mathbb{Z}/2$ -torsor $E_{\nu^*} \rightarrow E_{\nu^*}/P_{\nu^*} \cong E_{\nu}$. Let $\mathcal{E}_{\nu^*}$ denote the smooth locus of the closed subscheme of $\mathbb{P}_{\mathfrak{o}}^2$ cut out by the Legendre equation $y^2 = x(x-1)(x-\nu^*)$. Since $v(\nu^*) = 0$, the point P_{ν^*} spreads out to $\mathcal{P}_{\nu^*} \in \mathcal{E}_{\nu^*}(\mathfrak{o})$. The group structure of E_{ν^*} extends to $\mathcal{E}_{\nu^*}$ ([Sil94, Ch. IV, Theorem 5.3]), so we get another $\mathbb{Z}/2$ -torsor $\mathcal{E}_{\nu^*} \rightarrow \mathcal{E}_{\nu^*}/\mathcal{P}_{\nu^*}$, hence a class $\alpha_{P_{\nu^*}, \mathfrak{o}} \in H^1(\mathcal{E}_{\nu^*}/\mathcal{P}_{\nu^*}, \mathbb{Z}/2)$ whose image in $H^1(E_{\nu}, \mathbb{Z}/2)$ is $\alpha_{P_{\nu}}$. Since $v(\nu^* - 1) > 0$, the special fiber of $\mathcal{E}_{\nu^*} \rightarrow \mathcal{E}_{\nu^*}/\mathcal{P}_{\nu^*}$ is isomorphic to the double cover $\mathbb{G}_m \rightarrow \mathbb{G}_m$. Therefore, $\alpha_{P_{\nu^*}, \mathfrak{o}}$, the image of $\alpha_{P_{\nu^*}, \mathfrak{o}}$ under $\mathbb{Z}/2 \xrightarrow{\sim} \mu_2$, satisfies condition (iii) of Proposition 3.2 (for which we take $\mathcal{Y}_1 = \mathcal{E}_{\nu^*}/\mathcal{P}_{\nu^*}$) by the exact sequence

$$H^1(\kappa, \mu_2) \rightarrow H^1(\kappa(t), \mu_2) \xrightarrow{\prod_x r_{v_x}} \prod_x \mathbb{Z}/2,$$

where x ranges over the closed points of $\mathbb{P}_{\kappa}^1$ and v_x is the associated discrete valuation ([Ser97, II, Appendix, §4]).

Let $\mathcal{S}$ denote the smooth $\mathfrak{o}$ -model of S ; we will take $\mathcal{X} = \mathcal{S}$. We claim that, after finitely extending $\mathfrak{o}$, there exists $\beta_{\mathfrak{o}} \in H^2(\mathcal{S}, \mu_2)$ whose image in $H^2(S, \mu_2)$ is β and whose image

⁴For details on the splitting in the proof of *loc. cit.*, see MathOverflow questions 386147 and 462675.

in $H^2(\mathcal{S}_{\bar{\kappa}}, \mu_2)$ is not algebraic. Granted this, the image of $\beta_{\mathfrak{o}}$ is nonzero in $H^2(\kappa'(\mathcal{S}), \mu_2)$ for any field extension κ'/κ , because $\mathrm{Br}(\mathcal{S}_{\bar{\kappa}}) \rightarrow \mathrm{Br}(\mathcal{S}_{\kappa'})$ is an isomorphism when κ' is algebraically closed, so $\beta_{\mathfrak{o}}$ satisfies condition (ii) of Proposition 3.2. To find such a $\beta_{\mathfrak{o}}$, let $\bar{\mathfrak{o}}$ be a strict Henselization of $\mathfrak{o}$ inside k (cf. [Mil80, Ch. I, Example 4.10 and p. 38]), and let $f: \mathcal{S}_{\bar{\mathfrak{o}}} \rightarrow \mathrm{Spec}(\bar{\mathfrak{o}})$ be the obvious map. By the theorems on smooth and proper base change, the maps $\eta: H^2(\mathcal{S}_{\bar{\mathfrak{o}}}, \mu_{2^n}) \rightarrow H^2(S, \mu_{2^n})$ are isomorphisms (cf. the proof of [Fu15, Lemma 7.8.2]). Now $H^2(\mathcal{S}_{\bar{\mathfrak{o}}}, \mu_2)$ is the filtered colimit of $H^2(\mathcal{S}_{\mathfrak{o}'}, \mu_2)$ for certain finite ring maps $\mathfrak{o} \rightarrow \mathfrak{o}'$ ([Mil80, Ch. III, Lemma 1.16]); we choose $\beta_{\mathfrak{o}}$ by taking a preimage of β along η and descending to some $\mathfrak{o}'$ (and then replace $\mathfrak{o}$ with $\mathfrak{o}'$ and k_0 with the fraction field of $\mathfrak{o}'$). To show that the image of $\beta_{\mathfrak{o}}$ in $H^2(\mathcal{S}_{\bar{\kappa}}, \mu_2)$ is not algebraic, observe that the restriction maps $H^2(\mathcal{S}_{\bar{\mathfrak{o}}}, \mu_{2^n}) \rightarrow H^2(\mathcal{S}_{\bar{\kappa}}, \mu_{2^n})$ are also isomorphisms by the proper base change theorem ([Fu15, Corollary 7.3.3]). Thus, since being algebraic is the same as lifting to a class with μ_4 -coefficients (see (2.1)), and β is not algebraic, the same is true for the image of $\beta_{\mathfrak{o}}$. ■

For the notation and terminology in the following statement, see (1.5), (1.7), and (2.4).

3.4. Corollary. *Let k be an algebraically closed field of characteristic not 2. Let $\lambda, \mu, \nu \in k \setminus \{0, 1\}$, and let $\beta \in H^2(S_{\lambda, \mu}, \mu_2)$ be any non-algebraic class. Assume that (λ, μ, ν) is suitable. Then the image of $\beta \otimes_{\alpha_{P_\nu}}$ in $H^3(k(S_{\lambda, \mu} \times E_\nu), \mu_2^{\otimes 2})$ is nonzero.*

Proof. Case 1.5(a) is a special case of Corollary 3.3. In general, consider the pullback of $\beta \otimes_{\alpha_{P_\nu}}$ to $E_\lambda \times E_\mu \times E_\nu$. By Proposition 2.5, it equals $\alpha_{P_\lambda^\vee} \otimes \alpha_{P_\mu} \otimes \alpha_{P_\nu}$. We conclude the proof by using the same arguments as given above to apply Proposition 3.2. Specifically, in case 1.7(b), the assignment $(X, Y_1, Y_2, \xi, \alpha_1, \alpha_2) = (E_{\lambda_1}, E_{\lambda_2}, E_{\lambda_3}, \alpha_{P_{\lambda_1}^\vee}, \alpha_{P_{\lambda_2}}, \alpha_{P_{\lambda_3}})$ works, and in case 1.7(c), $(X, Y_1, \xi, \alpha_1) = (E_{\lambda_1} \times E_{\lambda_2}, E_{\lambda_3}, \alpha_{P_{\lambda_1}^\vee} \otimes \alpha_{P_{\lambda_2}}, \alpha_{P_{\lambda_3}})$ works. Actually, the argument simplifies, because in these cases $\xi_{\mathfrak{o}}$ can be written down explicitly. The clunky hypothesis “moreover there is no isogeny...” of 1.7(c) forces hypothesis (iii) of Proposition 3.2 to hold; more precisely, it guarantees that the class $\alpha_{P_{\lambda_1}^\vee} \otimes \alpha_{P_{\lambda_2}}$ spreads out to $\xi_{\mathfrak{o}}$ whose image in $H^2(E_{\bar{\lambda}_1} \times E_{\bar{\lambda}_2}, \mu_2)$ is not algebraic, hence nonzero in the Brauer group after any algebraically closed extension of the residue field (as in the proof of Corollary 3.3). ■

3.5. Remark. In some cases one can use an additional application of Proposition 3.2 to show that the isogeny condition of 1.7(c) holds, for example if there is a valuation on a subfield of the residue field of v with respect to which both $\bar{\lambda}_1$ and $\bar{\lambda}_2$ have positive valuation.

4. TORSION MOTIVES

Continue to assume that k is algebraically closed, and let p denote the exponential characteristic of k . For the definition of a Chow motive, see e.g. [And04, Ch. 3–4]. For a smooth projective variety X and a ring Λ , we write $\mathfrak{h}(X)_\Lambda$ for the Chow motive of X with coefficients in Λ , or simply $\mathfrak{h}(X)$ if $\Lambda = \mathbb{Z}$. We denote by $\mathbf{1} := \mathfrak{h}(\mathrm{Spec}(k))_\Lambda$ the unit object. We will use the usual “contravariant” normalization of the category of Chow motives, i.e. the functor from smooth projective varieties to Chow motives to be contravariant and $\mathfrak{h}(\mathbb{P}^1) \cong \mathbf{1} \oplus \mathbf{1}(-1)$.

4.1. Definition. A Chow motive M with coefficients in $\mathbb{Z}[1/a]$ is *torsion* if $(\text{End}(M), +)$ is a torsion group. If $t \cdot \text{End}(M) = 0$ for some positive integer t , we say that M is t -torsion, or simply that $t \cdot M = 0$. We call the smallest such t the *torsion order* of M . A torsion motive M always has a finite torsion order, namely, the order of id_M in $(\text{End}(M), +)$.

In the remainder of this section, we explain that two well-known facts hold in the setting of torsion motives. The first concerns torsion in CH^2 , following Merkurjev–Suslin ([MS82, §18]), and the second concerns the relationship between unramified cohomology and algebraicity of cohomology classes, following Colliot–Thélène–Voisin ([CTV12]). We remark that, for the purposes of this paper, we could make do with citing statements already in the literature. However, the “motivic” Propositions 4.2 and 4.3 will be useful in future work.

4.2. Proposition. *Let X be a smooth projective variety over k , let M be a torsion summand of $\mathfrak{h}(X)_{\mathbb{Z}[1/a]}$, and let $\ell \neq p$ be a prime. Then the cycle-class map $\text{cl}_{M,\ell}^i: \text{CH}^i(M)[\ell^\infty] \rightarrow H^{2i}(M, \mathbb{Z}_\ell(i))$ is bijective for $i = 1$ and injective for $i = 2$.*

Proof. By assumption, $\text{CH}^1(M)$ is a torsion summand of $\text{Pic}(X)$ of finite torsion order. Since $\text{Pic}^\circ(X) = \text{Ker}(\text{CH}^1(X) \rightarrow \text{NS}(X))$ is the group of points of an Abelian variety over an algebraically closed field, it has no direct summand of finite torsion order, so the map $\text{CH}^1(M) \rightarrow \text{NS}(X)$ is injective. Thus the map $\text{cl}_{M,\ell}^1: \text{CH}^1(M)[\ell^\infty] \rightarrow H^2(M, \mathbb{Z}_\ell(1))$ is injective as well. Now the Kummer sequence shows that $\text{Coker}(\text{cl}_{M,\ell}^1)$ embeds into $T_\ell(\text{Br}(X))$, the ℓ -adic Tate module of the Brauer group of X . But $T_\ell(\text{Br}(X))$ is torsion-free, so we must have $\text{Coker}(\text{cl}_{M,\ell}^1) = 0$ since M is torsion.

For the case $i = 2$, we will use Voevodsky’s Zariski (or, equivalently, Nisnevich) motivic cohomology $H_{\text{mot}}^{i,j}$. There is a functorial map $H_{\text{mot}}^{i,j}(X, \mathbb{Z}/\ell^n) \rightarrow H^i(X, \mu_{\ell^n}^{\otimes j})$ which, by the Beilinson–Lichtenbaum conjecture (now a theorem of Voevodsky), is injective if $i = j + 1$ (and an isomorphism if $i \leq j$); see [HW19, Corollary 1.31]. Letting $\mathbb{Z}_{(\ell)}$ denote the localization of $\mathbb{Z}$ at the prime ideal $\ell\mathbb{Z}$, there is a map $H_{\text{mot}}^{i,j}(X, \mathbb{Q}/\mathbb{Z}_{(\ell)}) \rightarrow H^i(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(j))$ because Zariski cohomology commutes with direct limits. Also, there is a map $f: H_{\text{mot}}^{i,j}(X, \mathbb{Z}_{(\ell)}) \rightarrow H^i(X, \mathbb{Z}_\ell(j))$ which is the inverse limit of compositions of the form $H_{\text{mot}}^{i,j}(X, \mathbb{Z}_{(\ell)}) \rightarrow H_{\text{mot}}^{i,j}(X, \mathbb{Z}/\ell^n) \rightarrow H^i(X, \mu_{\ell^n}^{\otimes j})$. We get a commutative diagram

$$(4.2.1) \quad \begin{array}{ccccc} H_{\text{mot}}^{3,2}(X, \mathbb{Q}/\mathbb{Z}_{(\ell)}) & \longrightarrow & H_{\text{mot}}^{4,2}(X, \mathbb{Z}_{(\ell)})_{\text{tors}} & \longrightarrow & 0 \\ \downarrow & & \downarrow f & & \\ H^3(X, \mathbb{Q}_\ell(2)) & \longrightarrow & H^3(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(2)) & \longrightarrow & H^4(X, \mathbb{Z}_\ell(2)) \end{array}$$

with exact rows coming from the short exact sequences $0 \rightarrow \mathbb{Z}_{(\ell)} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z}_{(\ell)} \rightarrow 0$ and $0 \rightarrow \mathbb{Z}_\ell \rightarrow \mathbb{Q}_\ell \rightarrow \mathbb{Q}_\ell/\mathbb{Z}_\ell \rightarrow 0$, respectively; the downward arrow on the left is injective by the Beilinson–Lichtenbaum conjecture. Moreover, f is the cycle-class map under the identification $H_{\text{mot}}^{2i,i}(X, \mathbb{Z}) \cong \text{CH}^i(X)$ of [Voe02, Corollary 2].

The functor from smooth projective varieties to Voevodsky’s triangulated category of motives factors through the category of Chow motives ([Voe02, Proposition 2.1.4]). In

particular, we obtain the diagram (4.2.1) with X replaced by M everywhere. The claim then follows from the fact that $H^3(M, \mathbb{Q}_\ell(2)) = 0$ because M is torsion. $\blacksquare$

4.3. Proposition. *Let M be a torsion summand of $\mathfrak{h}(X)_{\mathbb{Z}[1/a]}$, where X is a connected smooth projective variety over k . Then, for any prime $\ell \neq p$ and any $n \geq 1$, the boundary map $\partial: H^3(M, \mu_{\ell^n}^{\otimes 2}) \rightarrow H^4(M, \mathbb{Z}_\ell(2))$ induces an injective map*

$$(4.3.1) \quad \text{Img}\left(H^3(M, \mu_{\ell^n}^{\otimes 2}) \rightarrow H^3(k(X), \mu_{\ell^n}^{\otimes 2})\right) \hookrightarrow \text{Coker}(\text{cl}_{M,\ell}^2),$$

where $\text{cl}_{M,\ell}^2$ denotes the cycle-class map $\text{CH}^2(M) \rightarrow H^4(M, \mathbb{Z}_\ell(2))$.

Proof. The proof relies on the theory of unramified cohomology, which, according to [Sch23], may be defined as follows (though the specifics of the definition are not important here). For A being either $\mu_{\ell^n}^{\otimes j}$ or $\mathbb{Z}_\ell(j)$, we define $H^i(F_c X, A)$ to be the colimit of $H^i(U, A)$ as U ranges over open subsets of X of codimension $> c$, and then define $H_{\text{ur}}^i(X, A)$ to be the image of $H^i(F_1 X, A) \rightarrow H^i(F_0 X, A)$.⁵ We have maps $H^i(X, A) \rightarrow H_{\text{ur}}^i(X, A)$ and, for X integral, an injection $H_{\text{ur}}^i(X, \mu_{\ell^n}^{\otimes j}) \hookrightarrow H^i(k(X), \mu_{\ell^n}^{\otimes j})$ (indeed, the natural map $H^i(F_0 X, A) \rightarrow H^i(k(X), \mu_{\ell^n}^{\otimes j})$ is an isomorphism).

The link between unramified cohomology and non-algebraicity of cohomology classes is the existence of a commutative diagram with exact rows of the form

$$(4.3.2) \quad \begin{array}{ccccccc} H^3(X, \mu_{\ell^n}^{\otimes 2}) & \xrightarrow{\partial} & H^4(X, \mathbb{Z}_\ell(2)) & & & & \\ & \downarrow & & \downarrow & & & \\ H_{\text{ur}}^3(X, \mathbb{Z}_\ell(2)) & \longrightarrow & H_{\text{ur}}^3(X, \mu_{\ell^n}^{\otimes 2}) & \longrightarrow & \text{Coker}(\text{cl}_{X,\ell}^2) & \longrightarrow & 0, \end{array}$$

where ∂ is the usual boundary map in ℓ -adic cohomology. This was first proven by Colliot-Thélène and Voisin (for $k = \mathbb{C}$ and Betti cohomology; see [CTV12, Théorème 3.7]), and an “elementary” argument is explained in [Sch23, §3]. Note that Schreieder works with the exact sequence $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$ in Betti cohomology, but the same argument works for the exact sequence $0 \rightarrow \mathbb{Z}_\ell(2) \rightarrow \mathbb{Z}_\ell(2) \rightarrow \mu_{\ell^n}^{\otimes 2} \rightarrow 0$ in ℓ -adic cohomology. In particular, the map (4.3.1) exists.

A consequence of Bloch–Ogus’s proof of the Gersten conjecture for ℓ -adic cohomology ([BO74]) is that the assignment $Y \mapsto H_{\text{ur}}^i(Y, A)$ for smooth projective Y extends to a functor on the category of Chow motives; see [Sch25, Corollary 1.7] for a concrete (and more general) statement. Therefore, we obtain the diagram (4.3.2) with X replaced by M everywhere. By [Kah12, Lemme 3.12] (see also [CTV12, Théorème 3.1]), $H_{\text{ur}}^3(M, \mathbb{Z}_\ell(2))$ is torsion-free, hence trivial (because M is torsion). This proves the injectivity of (4.3.1). $\blacksquare$

⁵In general, Schreieder defines unramified cohomology in terms of a Borel–Moore homology theory; since X is smooth, this agrees with the definition above by Poincaré duality ([Sch23, §1.4]). An argument with the Gysin sequence shows that it also agrees with the usual definitions of unramified cohomology ([Sch23, equation (1.4)]).

5. TORSION IN THE MOTIVE OF A SURFACE

Continue to assume that k is algebraically closed of exponential characteristic p . In this section, we draw from Proposition 4.2 numerous consequences for torsion summands of the motives of surfaces. Then we explain the only known source of such torsion motives, namely surfaces admitting a decomposition of the diagonal (Proposition 5.6).⁶

If M is the summand of $\mathfrak{h}(X)$ cut out by a correspondence ε , let us denote by M^t the summand cut out by the transposed correspondence ε^t (not to be confused with $M^\vee := M^t(\dim(X))$, the usual duality functor on Chow motives.)

5.1. Let S be a smooth projective surface over k , and let M be a torsion summand of $\mathfrak{h}(S)_{\mathbb{Z}[1/a]}$. We will assume that the torsion order of M is a power of a prime ℓ different from p .⁷ By Roitman's theorem ([Roi80]), $\mathrm{CH}_0(M) = 0$. Thus $\mathrm{CH}^i(M) = 0$ for $i \neq 1$, and the map $\mathrm{CH}^1(M) \rightarrow H^2(M, \mathbb{Z}_\ell)$ is an isomorphism by Proposition 4.2. Also, $H^i(M, \mathbb{Z}_\ell) = 0$ for $i \neq 2, 3$. Poincaré duality induces isomorphisms $H^2(M, \mathbb{Z}_\ell)^\vee \cong H^3(M^t, \mathbb{Z}_\ell)$ and $H^3(M, \mathbb{Z}_\ell)^\vee \cong H^2(M^t, \mathbb{Z}_\ell)$, where $(-)^\vee := \mathrm{Hom}(-, \mathbb{Q}/\mathbb{Z})$; see (B.1)–(B.5) below.

5.2. Continuing in the setting of (5.1), let S' be another smooth projective surface over k , and let M' be a torsion summand of $\mathfrak{h}(S')_{\mathbb{Z}[1/a]}$, and assume that the torsion order of M' is also a power of ℓ . Then Proposition 4.2 and the Künneth theorem for ℓ -adic cohomology (see Proposition B.7 below) yield the following commutative diagram with exact rows, where we have abbreviated $H^i(-, \mathbb{Z}_\ell)$ by $H^i(-)$:

$$\begin{array}{ccccccc} 0 & \rightarrow & \mathrm{CH}^1(M^t) \otimes \mathrm{CH}^1(M') & \rightarrow & \mathrm{CH}^2(M^t \otimes M') & \longrightarrow & \mathrm{Coker} \longrightarrow 0 \\ & & \sim \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & H^2(M^t) \otimes H^2(M') & \longrightarrow & H^4(M^t \otimes M') & \longrightarrow & \bigoplus_{i=2,3} \mathrm{Hom}(H^i(M), H^i(M')) \longrightarrow 0. \end{array}$$

Note that $\mathrm{CH}^2(M^t \otimes M') = \mathrm{Hom}(M, M')$.

Moreover, the image of $\mathrm{Hom}(M, M')$ in $\bigoplus_i \mathrm{Hom}(H^i(M), H^i(M'))$ is dual to the image of $\mathrm{Hom}(M^t, M^t)$ in $\bigoplus_i \mathrm{Hom}(H^i(M^t), H^i(M^t))$ under Poincaré duality. More precisely, if $\varphi: M \rightarrow M'$ acts as (f_2, f_3) on cohomology, then the transpose φ^t defines a morphism $M^t \rightarrow M^t$, which acts as $(f_3^\vee, f_2^\vee)$ on cohomology.

5.3. Lemma. *Let M and M' be as in (5.2).*

- (a) *The torsion order M equals that of $H^*(M, \mathbb{Z}_\ell)$.*
- (b) *Suppose $\varphi: M \rightarrow M$ induces an automorphism of $H^*(M, \mathbb{Z}_\ell)$. Then φ is an automorphism.*

⁶In [Vis23a], Vishik constructs many other torsion objects in the stable motivic homotopy category, but these do not give rise to Chow motives, as noted in [Vis23a, Remark 2.2].

⁷Torsion motives have canonical “primary decompositions”, so we can take the “ ℓ -part” of any torsion summand of $\mathfrak{h}(S)_{\mathbb{Z}[1/a]}$.

- (c) Suppose $\varphi: M \rightarrow M'$ induces an isomorphism $H^*(M, \mathbb{Z}_\ell) \rightarrow H^*(M', \mathbb{Z}_\ell)$. Assume there exists some morphism $M' \rightarrow M$ which induces an isomorphism $H^*(M', \mathbb{Z}_\ell) \rightarrow H^*(M, \mathbb{Z}_\ell)$. Then φ is an isomorphism.
- (d) Assume that $M^t \cong M$ and $M^t \cong M'$, and suppose $\varphi: M \rightarrow M'$ induces an isomorphism $H^*(M, \mathbb{Z}_\ell) \rightarrow H^*(M', \mathbb{Z}_\ell)$. Then φ is an isomorphism.
- (e) Suppose $\varepsilon_1, \dots, \varepsilon_n: M \rightarrow M$ induce orthogonal idempotents in $\text{End}(H^*(M, \mathbb{Z}_\ell))$. Then there exist $\varepsilon'_1, \dots, \varepsilon'_n: M \rightarrow M$ which are themselves orthogonal idempotents in $\text{End}(M)$ such that ε'_i and ε_i induce the same endomorphism of $H^*(M, \mathbb{Z}_\ell)$ for each i .

Proof. (a). Obviously, the torsion order of M is a multiple of that of $H^*(M, \mathbb{Z}_\ell)$. On the other hand, the torsion order of M divides that of $H^4(M^t \otimes M, \mathbb{Z}_\ell)$ by Proposition 4.2. Since the Künneth sequence splits, the latter has torsion order equal to that of $H^*(M, \mathbb{Z}_\ell)$ by the isomorphism $H^2(M^t, \mathbb{Z}_\ell) \cong H^3(M, \mathbb{Z}_\ell)^\vee$ of Poincaré duality. This proves (a).

For any torsion summands N and N' of the motives of smooth projective surfaces, let $J(N, N') \subseteq \text{Hom}(N, N')$ denote the image of $\text{CH}^1(N^*) \otimes \text{CH}^1(N')$. Now let N'' be a third, and let $\varphi: N \rightarrow N'$ and $\varphi': N' \rightarrow N''$. One easily checks that $\varphi' \circ \varphi \in J(N, N'')$ if either $\varphi \in J(N, N')$ or $\varphi' \in J(N', N'')$ and that $\varphi' \circ \varphi = 0$ if $\varphi \in J(N, N')$ and $\varphi' \in J(N', N'')$.

(b). Since $H^*(M, \mathbb{Z}_\ell)$ is finite, some power φ^{or} of φ acts as the identity on $H^*(M, \mathbb{Z}_\ell)$, i.e. $\varphi^{\text{or}} = 1 + \varepsilon$ for some $\varepsilon \in J(M, M)$. Since $\varepsilon^2 = 0$, we have $(1 - \varepsilon) \circ \varphi^{\text{or}} = \varphi^{\text{or}} \circ (1 - \varepsilon) = 1$, so φ is an automorphism.

(c). This follows from (b) upon considering $\varphi \circ \psi$ and $\psi \circ \varphi$.

(d). The given isomorphisms allow us to view φ^t as an element of $\text{Hom}(M', M)$, and this map acts as an isomorphism on cohomology. So, the desired result follows from (c).

(e). The kernel of $\text{End}(M) \rightarrow \text{End}(H^*(M, \mathbb{Z}_\ell))$ is $J(M, M)$, a two-sided nilpotent ideal, so this follows from [Jan94, Lemma 5.4]. ■

5.4. Remark. We will frequently use the following special case of Lemma 5.3(c): if $S = S'$ and $H^*(M, \mathbb{Z}_\ell) = H^*(M', \mathbb{Z}_\ell)$ as summands of $H^*(S, \mathbb{Z}_\ell)$, then the compositions $M \rightarrow \mathfrak{h}(S)_{\mathbb{Z}[1/a]} \rightarrow M'$ and $M' \rightarrow \mathfrak{h}(S)_{\mathbb{Z}[1/a]} \rightarrow M$ are isomorphisms.

5.5. Remark. Lemma 5.3(c) is false without the assumption that there exists a morphism $M' \rightarrow M$ inducing an isomorphism on cohomology, as we will show in Example 6.7 below.

The following proposition, which provides many examples of torsion motives, is essentially [SY25, Proposition 3.4], which itself is a generalization of [GO13, Propositions 2.2 and 2.3].

5.6. Proposition. *Let S be a smooth projective surface over k . The following are equivalent:*

- (i) $\text{CH}_0(S_{k'}) \cong \mathbb{Z}$ for any algebraically closed field k'/k .⁸
- (ii) *There is a decomposition $\mathfrak{h}(S)_{\mathbb{Z}[1/p]} = \mathbf{1} \oplus \mathbf{1}(-1)^{\rho(S)} \oplus \mathbf{1}(-2) \oplus \mathfrak{t}(S)$ in which $\mathfrak{t}(S)$ is torsion, where $\rho(S)$ is the Picard number of S .*

If these conditions hold, then the following are true about $\mathfrak{t}(S)$.

- (a) $\mathfrak{t}(S)$ is well-defined up to isomorphism and satisfies $\mathfrak{t}(S)^t \cong \mathfrak{t}(S)$.

⁸See [KS16, Proposition 3.1.1] for several other equivalent conditions of the same flavor.

(b) We have $H^*(\mathfrak{t}(S), \mathbb{Z}_\ell) = H^*(S, \mathbb{Z}_\ell)_{\text{tors}}$ for any prime $\ell \neq p$ and $\text{CH}^*(\mathfrak{t}(S)) = \text{CH}^1(S)_{\text{tors}}$.

Proof. Assume (ii). Because the decomposition is stable under extension of the base field, for any algebraically closed field k'/k , we have $\text{CH}_0(S_{k'})_{\mathbb{Z}[1/p]} \cong \mathbb{Z}[1/p] \oplus A_{k'}$ for some group $A_{k'}$ of finite torsion order. This implies (i) by Roitman's theorem ([Roi80]) and its strengthening in positive characteristic due to Milne ([Mil82, Theorem 0.1]).

Now assume (i). Since the Albanese variety of S is trivial, the Picard scheme of S is zero-dimensional. In particular, the group of numerically trivial cycles in $\text{CH}^1(S)$ is precisely the torsion subgroup.

Pick $x \in S(k)$ and divisors $C_1, D_1, \dots, C_{\rho(S)}, D_{\rho(S)}$ on S for which $([C_1], \dots, [C_{\rho(S)}])$ and $([D_1], \dots, [D_{\rho(S)}])$ are dual bases of $\text{CH}^1(S)_{\mathbb{Q}}/\sim_{\text{num}}$ with respect to the intersection pairing. We get a decomposition $\mathfrak{h}(S)_{\mathbb{Q}} = \mathbf{1} \oplus \mathbf{1}(-1)^{\rho(S)} \oplus \mathbf{1}(-2) \oplus M$ in which the summands $\mathbf{1}$, $\mathbf{1}(-1)^{\rho(S)}$, and $\mathbf{1}(-2)$ are cut out by the projectors

$$\varepsilon^0 := \{x\} \times S, \quad \varepsilon^2 := \sum_{i=1}^{\rho(S)} [C_i \times D_i], \quad \text{and} \quad \varepsilon^4 := S \times \{x\},$$

respectively, and M is cut out by the “complementary” projector $[\Delta_S] - \varepsilon^0 - \varepsilon^2 - \varepsilon^4$. By construction, for any algebraically closed field k'/k , $\text{CH}^*(M_{k'})$ is the subgroup of numerically trivial cycles on $S_{k'}$. Thus, by (i) and the previous paragraph, $\text{CH}^*(M_{k'}) = 0$, so $M = 0$ by [GG12, Lemma 2.1]. We deduce that $\rho(S) = b_2(S)$, and therefore that the map $\text{NS}(S) \otimes \mathbb{Z}_\ell \rightarrow H^2(S, \mathbb{Z}_\ell(1))$ is an isomorphism for all primes $\ell \neq p$. (The cokernel is, by the Kummer sequence, the Tate module $T_\ell(\text{Br}(S))$, which is torsion-free.) By Poincaré duality for ℓ -adic cohomology, the intersection pairing on $\text{CH}^1(S)_{\mathbb{Z}[1/p]}/\sim_{\text{num}}$ is perfect. Therefore, we may assume that the C_i and D_i above form dual bases of $\text{CH}^1(S)_{\mathbb{Z}[1/p]}/\sim_{\text{num}}$. Then we obtain the decomposition described in (ii). Statement (b) is clear. Statement (a) follows from Remark 5.4 (but can also be proven directly at the level of cycles). ■

5.7. If $\text{char}(k) = 0$ and S is a surface over k satisfying $b_1(S) = 0$ and $b_2(S) = \rho(S)$, then we expect that $\text{CH}_0(S) \cong \mathbb{Z}$ according to a classical conjecture of Bloch ([Blo75]). In a recent preprint ([Gul25, Theorems A and B]), Guletskiĭ announced a proof of this conjecture, together with the following positive-characteristic version: If $\text{char}(k) > 0$, S has a model over $\overline{\mathbb{F}_p}$, $b_1(S) = 0$, and $b_2(S) = \rho(S)$, then $\text{CH}_0(S) \cong \mathbb{Z}$. Previously, Bloch's conjecture was proven for all surfaces not of general type ([BKL76]) and several classes of surfaces of general type (due to many authors, starting with [IM79]; see e.g. the references in [PW16] and [Gul25]).

5.8. Remark. In [SY25, Proposition 3.6(1)] (see also [SY25, Proposition 2.14(2)]), it is incorrectly stated that the map $\text{CH}^2(M^t \otimes M') \rightarrow \bigoplus_{i=2,3} \text{Hom}(H^i(M), H^i(M'))$ is injective whenever S and S' are as in Proposition 5.6. The issue is that the first isomorphism written on the top of page 14 of *loc. cit.* is false in general, because some of the torsion in the cohomology of a product is in the image of the Künneth map. In particular, the answer to [SY25, Problem 3.8] is “no”. This does not effect the other results of *loc. cit.* (but “eff” must be replaced by “nor” in the exact sequence [SY25, equation (7.4)]).

6. THE MOTIVE OF AN ENRIQUES SURFACE

Continue to assume that k is algebraically closed, and let p denote the exponential characteristic of k , but now additionally assume that $p \neq 2$.

6.1. Let S be an Enriques surface over k . Recall that the 2-adic cohomology groups of S are $(\mathbb{Z}_2, 0, \mathbb{Z}_2^{10} \oplus \mathbb{Z}/2, \mathbb{Z}/2, \mathbb{Z}_2)$, and the ℓ -adic cohomology of S is torsion-free for any prime $\ell \nmid 2p$ ([CDL25, §1.4]). By [BKL76], the equivalent conditions of Proposition 5.6 hold for S ; in particular, the torsion summand $\mathfrak{t}(S)$ of $\mathfrak{h}(S)_{\mathbb{Z}[1/p]}$ exists, and it is 2-torsion by Lemma 5.3(a). Now let S' be another Enriques surface over k . Then (5.2) yields a split exact sequence

$$0 \rightarrow \mathbb{Z}/2 \rightarrow \mathrm{Hom}(\mathfrak{t}(S), \mathfrak{t}(S')) \xrightarrow{\tau} \bigoplus_{i=2}^3 \mathrm{Hom}(H^i(S, \mathbb{Z}_2)_{\mathrm{tors}}, H^i(S', \mathbb{Z}_2)_{\mathrm{tors}}),$$

and the last term can be identified with $\mathbb{Z}/2 \oplus \mathbb{Z}/2$. Let $I(S, S')$ denote the image of τ , viewed as a subgroup $\mathbb{Z}/2 \oplus \mathbb{Z}/2$. Notice that $\mathrm{Coker}(\tau) = H^4(S \times S', \mathbb{Z}_2)_{\mathrm{tors}} / (\text{algebraic classes})$.

6.2. Lemma. *Let S and S' be Enriques surfaces over k . In the notation of (6.1), we have that*

- (a) $\mathfrak{t}(S) \cong \mathfrak{t}(S')$ if and only if $(1, 1) \in I(S, S')$.
- (b) $\mathfrak{t}(S)$ is indecomposable if and only if $I(S, S') = \{(0, 0), (1, 1)\}$.
- (c) $\mathfrak{t}(S)$ is decomposable if and only if $I(S, S') = \mathbb{Z}/2 \oplus \mathbb{Z}/2$. If this is the case, $\mathfrak{t}(S) = \mathfrak{t}^2(S) \oplus \mathfrak{t}^3(S)$ for nontrivial motives $\mathfrak{t}^2(S)$ and $\mathfrak{t}^3(S)$ which are unique up to isomorphism. They satisfy $H^*(\mathfrak{t}^i(S), \mathbb{Z}_2) = H^i(S, \mathbb{Z}_2)_{\mathrm{tors}}$, $\mathrm{CH}^*(\mathfrak{t}^2(S)) = \mathrm{CH}^1(S)_{\mathrm{tors}}$, $\mathrm{CH}^*(\mathfrak{t}^3(S)) = 0$, and $\mathfrak{t}^2(S)^{\mathfrak{t}} \cong \mathfrak{t}^3(S)$.

Proof. Statement (a) follows from Lemma 5.3(d). Noting that $(1, 1) = \tau(\mathrm{id}_{\mathfrak{t}(S)}) \in I(S, S)$, statement (b) and the first sentence of (c) follow from Lemma 5.3(e). The computation of $H^*(\mathfrak{t}^i(S), \mathbb{Z}_2)$ and $\mathrm{CH}^*(\mathfrak{t}^i(S))$ is easy, and the former implies the uniqueness of $\mathfrak{t}^2(S)$ and $\mathfrak{t}^3(S)$ as well as the isomorphism $\mathfrak{t}^2(S)^{\mathfrak{t}} \cong \mathfrak{t}^3(S)$ by Remark 5.4. ■

6.3. Remark. Since $\mathrm{CH}^2(S_{k(S')})$ is the direct limit of $\mathrm{CH}^2(S \times U)$, where U ranges over the nonempty open subvarieties of S' , we have $I(S, S') \cong \mathrm{CH}_0(S_{k(S')})$ by the localization sequence for Chow groups. In particular, if $\mathfrak{t}(S)$ is decomposable, then $\mathrm{CH}_0(S_{k(S)}) = \mathbb{Z}/2 \oplus \mathbb{Z}/2$. In turn, $H_{\mathrm{ur}}^3(S \times S, \mathbb{Q}_{\ell}/\mathbb{Z}_{\ell}(2)) = 0$ for any prime $\ell \neq p$ by [Kah17, Corollary 6.4(a)].

We will show in Examples 6.5 and 6.6 that, when $\mathfrak{t}(S) \not\cong \mathfrak{t}(S')$, all three possibilities for $I(S, S')$ do occur. For now, we turn toward the proofs of Theorems 1.2 and 1.8. The former will be a consequence of the following more general fact.

6.4. Proposition. *Let $f: S \dashrightarrow S'$ be a rational map between Enriques surfaces over k .*

- (a) *Assume f is of odd degree. Then $\mathfrak{t}(S) \cong \mathfrak{t}(S')$.*
- (b) *Assume f is of even degree. Then $(0, 1) \in I(S, S')$. In particular, if $S = S'$, then $\mathfrak{t}(S)$ is decomposable.*

Proof. Resolving f gives a diagram $S \xleftarrow{\pi} \tilde{S} \xrightarrow{\tilde{f}} S'$ in which π is a composition of blowups and $\tilde{f}$ is a generically finite morphism of odd degree. By the blowup formula (2.2), the map

$\pi^*: H^*(S, \mathbb{Z}_2)_{\text{tors}} \rightarrow H^*(\tilde{S}, \mathbb{Z}_2)_{\text{tors}}$ is an isomorphism, and by Poincaré duality, so is π_* . In particular, $H^i(\tilde{S}, \mathbb{Z}_2)_{\text{tors}} \cong \mathbb{Z}/2$ for $i \in \{2, 3\}$.

(a). Since $\tilde{f}$ is generically finite of odd degree, the projection formula shows that $\tilde{f}_* \circ \tilde{f}^*$ acts on cohomology as multiplication by an odd number. We deduce that $\tilde{f}_* \circ \pi^*$ induces an isomorphism $H^*(S, \mathbb{Z}_2)_{\text{tors}} \xrightarrow{\sim} H^*(S', \mathbb{Z}_2)_{\text{tors}}$. Therefore, the image of $\tilde{S}$ in $S \times S'$ defines an element of $\text{CH}^2(S \times S')$ whose image in $\text{CH}^2(\mathfrak{t}(S)^\vee \otimes \mathfrak{t}(S'))$ induces an isomorphism $H^*(\mathfrak{t}(S), \mathbb{Z}_2) \xrightarrow{\sim} H^*(\mathfrak{t}(S'), \mathbb{Z}_2)$. We conclude the proof by applying Lemma 5.3(d).

(b). Consider the fiber product

$$\begin{array}{ccc} \tilde{X} & \longrightarrow & X' \\ \downarrow & & \downarrow \\ \tilde{S} & \xrightarrow{\tilde{f}} & S', \end{array}$$

where $X' \rightarrow S'$ is K3 cover. If $\tilde{X}$ were disconnected, then $\tilde{f}$ would factor through a surjective morphism $\tilde{S} \rightarrow X'$; but this is impossible, because $\tilde{S}$ would inherit a nontrivial holomorphic 2-form from X . Therefore, $\tilde{f}^*: H^1(S', \mathbb{Z}/2) \rightarrow H^1(\tilde{S}, \mathbb{Z}/2)$ is injective. Since $H^1(S', \mathbb{Z}_2) \cong H^1(\tilde{S}, \mathbb{Z}_2) \cong 0$ by the blowup formula, the boundary morphisms $H^1(S', \mathbb{Z}/2) \rightarrow H^2(S', \mathbb{Z}_2)_{\text{tors}}$ and $H^1(\tilde{S}, \mathbb{Z}/2) \rightarrow H^2(\tilde{S}, \mathbb{Z}_2)_{\text{tors}}$ are isomorphisms. Thus $\tilde{f}^*: H^2(S', \mathbb{Z}_2)_{\text{tors}} \rightarrow H^2(\tilde{S}, \mathbb{Z}_2)_{\text{tors}}$ is also an isomorphism. By Poincaré duality, so is $\tilde{f}_*: H^3(\tilde{S}, \mathbb{Z}_2)_{\text{tors}} \rightarrow H^3(S', \mathbb{Z}_2)_{\text{tors}}$. Moreover, $\tilde{f}_* \circ \tilde{f}^*$ induces multiplication by an even number (hence zero) on $H^2(S', \mathbb{Z}_2)_{\text{tors}}$, so the map $\tilde{f}_*: H^2(\tilde{S}, \mathbb{Z}_2)_{\text{tors}} \rightarrow H^2(S', \mathbb{Z}_2)_{\text{tors}}$ must vanish. All in all, $\tilde{f}_* \circ \pi^*$ kills $H^2(S, \mathbb{Z}_2)_{\text{tors}}$ but acts as an isomorphism on $H^3(S', \mathbb{Z}_2)_{\text{tors}}$. We conclude, as in the proof of (a), by considering the image of $\tilde{S}$ in $S \times S'$; the statement for $S = S'$ follows from Lemma 5.3(e). $\blacksquare$

Proof of Theorem 1.2. Let $\circ, \bullet \in \mathfrak{L}$. If $\varphi: E_\circ \rightarrow E_\bullet$ and $\psi: F_\circ \rightarrow F_\bullet$ are isogenies satisfying $\varphi(P_\circ) = P_\bullet$ and $\psi(Q_\circ) = Q_\bullet$, then φ and ψ induce a rational map $S_\circ \dashrightarrow S_\bullet$ of degree $\deg(\varphi) \cdot \deg(\psi)$. Therefore, statement (a) of the present theorem follows immediately from Proposition 6.4(a).

(b). Recall that $E_\circ$ is an elliptic curve over k such that $\text{End}(E_\circ) \cong \mathcal{O}_L$, where L is a quadratic imaginary number field in which 2 splits. By the previous paragraph and Proposition 6.4(b), it suffices to find $P_\circ \in E_\circ[2] \setminus \{O\}$ and an endomorphism $\varphi \in \text{End}(E_\circ)$ of even degree satisfying $\varphi(P_\circ) = P_\circ$. By hypothesis, $E_\circ[2]$ is isomorphic, as an $\text{End}(E_\circ) \cong \mathcal{O}_L$ -module, to $\mathcal{O}_L/2 \cong \mathcal{O}_L/\mathfrak{p} \times \mathcal{O}_L/\bar{\mathfrak{p}} \cong \mathbb{F}_2 \times \mathbb{F}_2$, where $2\mathcal{O}_L = \mathfrak{p} \cdot \bar{\mathfrak{p}}$ is the factorization into prime ideals. Thus, if we pick $P_\circ$ and φ whose images in $\mathbb{F}_2 \times \mathbb{F}_2$ (under the maps

$$E_\circ[2] \cong \mathcal{O}_L/2 \cong \mathbb{F}_2 \times \mathbb{F}_2 \quad \text{and} \quad \text{End}(E_\circ) \cong \mathcal{O}_L \twoheadrightarrow \mathcal{O}_L/2 \cong \mathbb{F}_2 \times \mathbb{F}_2,$$

respectively) coincide and equal $(1, 0)$ or $(0, 1)$, then $\deg(\varphi)$ is even and $\varphi(P_\circ) = P_\circ$. $\blacksquare$

Proof of Theorem 1.8. Let's assume that $(\lambda, \mu, \lambda'^*)$ is suitable; the other cases are similar. Let α and α' denote the unique nonzero elements in $H^1(S_{\lambda, \mu}, \mu_2)$ and $H^1(S_{\lambda', \mu'}, \mu_2)$, respectively, and let $\alpha_{P_{\lambda'^*}} \in H^1(E_{\lambda'^*}, \mu_2)$ be as in 2.4(b). Note that $H^1(\mathfrak{t}(S_{\lambda, \mu}), \mu_2) = H^1(S_{\lambda, \mu}, \mu_2)$, and similarly for $S_{\lambda', \mu'}$ (and also for H^3).

There exists a pullback diagram of the form

$$\begin{array}{ccc} E_{\lambda'} & \longrightarrow & \text{Kum}(E_{\lambda'} \times E_{\mu'}) \\ \downarrow & & \downarrow \\ E_{\lambda'}/P_{\lambda'} & \xrightarrow{f} & S_{\lambda',\mu'}. \end{array}$$

(This f is the inclusion of a half-fiber of an elliptic fibration on $S_{\lambda',\mu'}$.) As noted in (1.6), we have $E_{\lambda'}/P_{\lambda'} \cong E_{\lambda'^*}$ and, under this isomorphism, $f^*\alpha' = \alpha_{P_{\lambda'^*}}$. In particular, f induces a map $\mathfrak{t}(S_{\lambda',\mu'}) \rightarrow \mathfrak{h}(E_{\lambda'^*})$ sending α' to $\alpha_{P_{\lambda'^*}}$.

Now let β be any non-algebraic class in $H^2(S_{\lambda,\mu}, \mu_2)$, and let γ be the unique nonzero class in $H^3(S_{\lambda,\mu}, \mu_2)$. Then $\gamma = \bar{\partial}\beta$, where $\bar{\partial}$ is the boundary map $H^2(S_{\lambda,\mu}, \mu_2) \rightarrow H^3(S_{\lambda,\mu}, \mu_2)$, i.e. the composition of the boundary map $\partial: H^2(S_{\lambda,\mu}, \mu_2) \rightarrow H^3(S_{\lambda,\mu}, \mathbb{Z}_2(1))$ with reduction mod 2. Since $\partial\alpha_{P_{\lambda'^*}} = 0$, the Leibniz rule gives $\bar{\partial}(\beta \otimes \alpha_{P_{\lambda'^*}}) = \gamma \otimes \alpha_{P_{\lambda'^*}}$. Thus the class $\gamma \otimes \alpha_{P_{\lambda'^*}} \in H^4(S_{\lambda,\mu} \times E_{\lambda'^*}, \mu_2^{\otimes 2})$ is not algebraic by Corollary 3.4, Proposition 4.3, and the assumption that (λ, μ, λ') is suitable. Equivalently, no correspondence $\Gamma \in \text{CH}^2(S_{\lambda,\mu} \times E_{\lambda'^*})$ satisfies $\Gamma_*\alpha = \alpha_{P_{\lambda'^*}}$ (see (B.6.1)). Therefore, there is no map $\mathfrak{t}(S_{\lambda,\mu}) \rightarrow \mathfrak{h}(E_{\lambda'^*})$ sending α to $\alpha_{P_{\lambda'^*}}$. Comparing with the final sentence of the previous paragraph, we see that $\mathfrak{t}(S_{\lambda,\mu}) \not\cong \mathfrak{t}(S_{\lambda',\mu'})$, as desired. $\blacksquare$

6.5. Example. Let $\lambda, \mu, \lambda', \mu' \in k \setminus \{0, 1\}$. Using that the boundary map $H^1(\mathfrak{t}(S), \mathbb{Z}/2) \rightarrow H^2(\mathfrak{t}(S), \mathbb{Z}_2)$ is an isomorphism for any Enriques surface S over k , it follows from the proof of Theorem 1.2 that if either $(\lambda, \mu, \lambda'^*)$ or (λ, μ, μ'^*) is suitable, then $I(S_{\lambda,\mu}, S_{\lambda',\mu'}) \subseteq 0 \oplus \mathbb{Z}/2$ and by Poincaré duality, $I(S_{\lambda',\mu'}, S_{\lambda,\mu}) \subseteq \mathbb{Z}/2 \oplus 0$. Therefore, if additionally one of $(\lambda', \mu', \lambda^*)$ or (λ', μ', μ^*) is suitable, then $I(S_{\lambda,\mu}, S_{\lambda',\mu'}) = 0$. For example, if $k = \overline{\mathbb{Q}}$, $\lambda = \mu = 25$, and $\lambda' = \mu' = 25^* = 9/4$, then 1.7(b) holds for $(\lambda, \mu, \lambda'^*)$ with $k_0 = \mathbb{Q}$ and v the 5-adic valuation and for $(\lambda', \mu', \lambda^*)$ with $k_0 = \mathbb{Q}$ and v the 3-adic valuation.

6.6. Example. Let $\lambda, \mu \in k \setminus \{0, 1\}$. Under the obvious isomorphism $E_{1-\lambda} \xrightarrow{\sim} E_{\lambda}$, the point $P_{1-\lambda}$ does not map to P_{λ} . That way, by (1.6), there is a degree-2 isogeny $E_{1-\lambda} \rightarrow E_{\lambda^*}$ which sends $P_{1-\lambda}$ to P_{λ^*} . As in the proof of Theorem 1.2, we obtain a degree-2 rational map $S_{1-\lambda,\mu} \dashrightarrow S_{\lambda^*,\mu}$. If, in addition, $(1-\lambda, \mu, \lambda)$ is suitable, then $I(S_{1-\lambda,\mu}, S_{\lambda^*,\mu}) = 0 \oplus \mathbb{Z}/2$ and $I(S_{\lambda^*,\mu}, S_{1-\lambda,\mu}) = \mathbb{Z}/2 \oplus 0$ by the previous paragraph and part (b) of Proposition 6.4. For example, if $k = \overline{\mathbb{Q}}$, $\lambda = 4/9$ (so $\lambda^* = 1/25$), and $\mu = -1$, then 1.7(b) holds for $(1-\lambda, \mu, \lambda)$ with $k_0 = \mathbb{Q}$ and v the 3-adic valuation.

6.7. Example. Assume that S and S' are Enriques surfaces over k such that $I(S, S') = 0 \oplus \mathbb{Z}/2$ and $\mathfrak{t}(S)$ and $\mathfrak{t}(S')$ are decomposable. Then the nonzero map $\mathfrak{t}(S) \rightarrow \mathfrak{t}(S')$ induces a map $\mathfrak{t}^3(S) \rightarrow \mathfrak{t}^3(S')$ which is an isomorphism on cohomology. However, it is not possible that $\mathfrak{t}^3(S) \cong \mathfrak{t}^3(S')$, for this would imply that $\mathfrak{t}^2(S) \cong \mathfrak{t}^2(S')$ (since $\mathfrak{t}^2$ is dual to $\mathfrak{t}^3$), hence in turn that $\mathfrak{t}(S) \cong \mathfrak{t}(S')$ (since $\mathfrak{t} = \mathfrak{t}^2 \oplus \mathfrak{t}^3$).

Explicitly, consider the curve E over $\overline{\mathbb{Q}}$ given by $y^2 = (x+7)(x^2-7x+14)$, which satisfies $\text{End}(E) \cong \mathcal{O}_{\mathbb{Q}(\sqrt{-7})}$ ([lmf26, elliptic curve 784.f4]). Let $\mu := (e_2+7)/(e_3+7)$, where e_2 and e_3 are the roots of $x^2-7x+14$; then $E \cong E_{\mu}$. Since e_2 and e_3 are conjugate over

$\mathbb{Q}$ and 3 is inert in $\mathbb{Q}(\sqrt{-7})$, we have $v(\mu) = 0$ for the 3-adic valuation v on $\mathbb{Q}(\sqrt{-7})$. Therefore $I(S_{-5/9,\mu}, S_{1/25,\mu}) = 0 \oplus \mathbb{Z}/2$ as in Example 6.6, while $\mathfrak{t}(S_{-5/9,\mu})$ and $\mathfrak{t}(S_{1/25,\mu})$ are decomposable by part (b) of Theorem 1.2.

6.8. Example. We can also construct examples where the Enriques surfaces share the same K3 cover. For example, if $\lambda, \mu \in k \setminus \{0, 1\}$ and $(\lambda, 1 - \mu, \mu^*)$ and $(1 - \lambda, \mu, \lambda^*)$ are both suitable, then $I(S_{\lambda, 1-\mu}, S_{1-\lambda, \mu}) = 0$, while $S_{\lambda, 1-\mu}$ and $S_{1-\lambda, \mu}$ have isomorphic K3 covers (since $E_\lambda \cong E_{1-\lambda}$ and $E_\mu \cong E_{1-\mu}$). Explicitly, if we take $k = \overline{\mathbb{Q}}$, $\lambda = 25$ (so $\lambda^* = 9/4$), and $\mu = 36$ (so $\mu^* = 49/25$), then $(\lambda, 1 - \mu, \mu^*)$ satisfies 1.7(b) with v the 5-adic valuation on $\mathbb{Q}$, and $(1 - \lambda, \mu, \lambda^*)$ satisfies 1.7(b) with v the 3-adic valuation on $\mathbb{Q}$.

APPENDIX A. ON [Vis23b, Proposition 4.21]

The error in the proof of [Vis23b, Proposition 4.21] arises in the phrase “this pair of Rost submodules satisfies all the conditions of Theorem 4.20(2)”. In fact, condition (a) of [Vis23b, Theorem 4.20(2)] fails, as we will explain in (A.5) below.

Let k be an algebraically closed field of characteristic 0.

A.1. For a ring R , let $\mathbf{DM}(k, R)$ denote Voevodsky’s triangulated category of motives over k with coefficients in R , and let $\mathbf{DM}_{\text{gm}}(k, R)$ be the subcategory of geometric motives. For $M \in \mathbf{DM}(k, R)$, its motivic cohomology and homology are given by $H_{\text{mot}}^{i,j}(M, R) := \text{Hom}_{\mathbf{DM}(k, R)}(M, R(j)[i])$ and $H_{i,j}^{\text{mot}}(M, R) := \text{Hom}_{\mathbf{DM}(k, R)}(R(j)[i], M)$, respectively (where, importantly, we take morphisms in the category with R -coefficients).

A.2. For any integer $n \geq 0$, the functor from the category of smooth projective k -varieties to $\mathbf{DM}(k, \mathbb{Z}/n)$ factors through a fully faithful functor $\mathbf{CHM}(k, \mathbb{Z}/n)^{\text{op}} \rightarrow \mathbf{DM}_{\text{gm}}(k, \mathbb{Z}/n)$. This is well-known when $n = 0$ (e.g. [MVW06, Proposition 20.1]); for $n \geq 1$, the same proof works once one checks that $H_{\text{mot}}^{2i,i}(M(X), \mathbb{Z}/n) \cong \text{CH}^i(X) \otimes \mathbb{Z}/n$, which can be done using the following adjoint triple of functors $\nu^* \dashv \nu_* \dashv \nu^*[-1]$:

$$\mathbf{DM}(k, \mathbb{Z}) \xrightarrow{\nu^*} \mathbf{DM}(k, \mathbb{Z}/n) \xrightarrow{\nu_*} \mathbf{DM}(k, \mathbb{Z}) \xrightarrow{\nu^*[-1]} \mathbf{DM}(k, \mathbb{Z}/n),$$

where $\nu^*(M) := \text{Cone}(n: M \rightarrow M)$ and ν_* views a motive with $\mathbb{Z}/n$ -coefficients as a motive with $\mathbb{Z}$ -coefficients.

The following application of Bondarko’s theory of weight structures ([Bon10]) allows one to detect the Chow motives in $\mathbf{DM}_{\text{gm}}(k, \mathbb{Z}/n)$ using motivic cohomology and homology.

A.3. Proposition. *Fix $n \geq 0$, and let $M \in \mathbf{DM}_{\text{gm}}(k, \mathbb{Z}/n)$. Then the following are equivalent:*

- (a) *M is a Chow motive with $\mathbb{Z}/n$ -coefficients, i.e. lives in the essential image of the functor $\mathbf{CHM}(k, \mathbb{Z}/n) \rightarrow \mathbf{DM}_{\text{gm}}(k, \mathbb{Z}/n)$.*
- (b) *For any finitely generated field extension k'/k , we have $H_{\text{mot}}^{i,j}(M_{k'}, \mathbb{Z}/n) = 0$ for all $i > 2j$, and $H_{i,j}^{\text{mot}}(M_{k'}, \mathbb{Z}/n) = 0$ for all $i < 2j$.*

Proof. When $n = 0$, this is [Vis23b, Proposition 5.1]. The same proof works for $n \geq 1$, using Bondarko’s Chow weight structure on $\mathbf{DM}_{\text{gm}}(k, \mathbb{Z}/n)$ instead ([Bon10, Remark 6.6.1]). ■

A.4. Corollary. *If $n \geq 1$, then no nontrivial Chow motive with $\mathbb{Z}$ -coefficients is of the form $\nu_*(N)$ for some $N \in \mathbf{DM}_{\text{gm}}(k, \mathbb{Z}/n)$.*

Proof. Let N be as above, and assume that its image $M := \nu_*(N)$ in $\mathbf{DM}(k, \mathbb{Z})$ is a Chow motive (with $\mathbb{Z}$ -coefficients). Using the adjoint triple of functors $\nu^* \dashv \nu_* \dashv \nu^*[-1]$ of (A.2), we find that $H_{i,j}^{\text{mot}}(N, \mathbb{Z}/n) \cong H_{i,j}^{\text{mot}}(M, \mathbb{Z})$ and $H_{\text{mot}}^{i,j}(N, \mathbb{Z}/n) \cong H_{\text{mot}}^{i+1,j}(M, \mathbb{Z})$ for all i, j . Thus, since M is a Chow motive, N is also a Chow motive (with $\mathbb{Z}/n$ -coefficients) by Proposition A.3. But also $\text{CH}^i(N) \cong H_{\text{mot}}^{2i,i}(N, \mathbb{Z}/n) \cong H_{\text{mot}}^{2i+1,i}(M, \mathbb{Z}) = 0$ for all i , and varying the base field k , the same argument gives that $\text{CH}^i(N_{k'}) = 0$ for all i and any field extension k'/k . This implies that $N = 0$, hence $M = 0$. $\blacksquare$

A.5. Now we explain in detail what goes wrong in [Vis23b, Proposition 4.21]. In the notation there, we claim that $N_\bullet(C)$ is not compact, i.e. not a geometric motive ([CD19, Theorem 11.1.13]), contrary to condition (a) of [Vis23b, Theorem 4.20(2)]. Indeed, if it were a geometric motive, the proof of [Vis23b, Theorem 4.20] would produce an exact triangle

$$\mathcal{T}_C(-1) \rightarrow N \rightarrow N_\bullet(C) \xrightarrow{+1}$$

in which $N \in \mathbf{DM}_{\text{gm}}(k, \mathbb{Z}/n)$ (and this N appears in the motive of a smooth projective curve). Moreover, tracing through the proof shows that application of ν_* to this exact triangle yields the exact triangle

$$\mathcal{T}_A(-1) \rightarrow M \rightarrow M_\bullet(A) \xrightarrow{+1}$$

defining $M_\bullet(A)$. (Note that $\nu_*(N_\bullet(C)) = M_\bullet(A)$ and $\nu_*(\mathcal{T}_C) = \mathcal{T}_A$ by definition, and similarly for the motives $\mathcal{K}_C$ and $\mathcal{K}_A$ used in [Vis23b].) We deduce from Corollary A.4 that $M = 0$, contrary to the hypothesis of [Vis23b, Proposition 4.21].

APPENDIX B. POINCARÉ AND KÜNNETH IN THE PRESENCE OF TORSION

Throughout, k is an algebraically closed field, X and Y are connected smooth projective varieties over k of respective dimensions d_X and d_Y , and ℓ is a fixed prime number not equal to $\text{char}(k)$. For simplicity, we use a fixed compatible system of roots of unity to identify all Tate twists in étale cohomology. However, everything can be adapted to work for general k , in which case all the maps (with the appropriate twists) become equivariant for the Galois action. Also, everything below has analogues for Betti and crystalline cohomology, and in the latter case, everything respects the Frobenius actions. The material in this section is well-known to experts, but we could not find some of it, especially Proposition B.7, in the literature.

We begin with some basic applications of Poincaré duality in étale cohomology.

B.1. Fix a positive integer n . There exists a “trace” isomorphism $\text{tr}_X: H^{2d_X}(X, \mathbb{Z}/\ell^n) \xrightarrow{\sim} \mathbb{Z}/\ell^n$ which makes the cup-product pairing

$$\langle -, - \rangle_n: H^i(X, \mathbb{Z}/\ell^n) \times H^{2d_X-i}(X, \mathbb{Z}/\ell^n) \rightarrow H^{2d_X}(X, \mathbb{Z}/\ell^n) \xrightarrow{\sim} \mathbb{Z}/\ell^n$$

perfect for each i ; this is a version of Poincaré duality for X in étale cohomology. Therefore, for a morphism $f: X \rightarrow Y$, we have a “pushforward” map $f_*: H^i(X, \mathbb{Z}/\ell^n) \rightarrow$

$H^{i+2(d_Y-d_X)}(Y, \mathbb{Z}/\ell^n)$ which is defined to be equal, under Poincaré duality, to $(f^*)^\vee$, where $(-)^\vee$ denotes the functor $\text{Hom}(-, \mathbb{Z}/\ell^n)$. In other words, f_* is defined by the identity $\langle f_*\alpha, \beta \rangle_n = \langle \alpha, f^*\beta \rangle_n$. Then, for any $\Gamma \in H^{2d_X+e}(X \times Y, \mathbb{Z}/\ell^n)$, we define $\Gamma_*: H^i(X, \mathbb{Z}/\ell^n) \rightarrow H^{i+e}(Y, \mathbb{Z}/\ell^n)$ by $\Gamma_*\alpha := \text{pr}_{2*}(\text{pr}_1^*\alpha \cdot \Gamma)$, where pr_1 and pr_2 denote the projection maps $X \times Y \rightarrow X$ and $X \times Y \rightarrow Y$, respectively.

B.2. Everything in the previous paragraph is compatible with the reduction maps $\mathbb{Z}/\ell^{n+1} \rightarrow \mathbb{Z}/\ell^n$, so we can define pushforward maps on $H^*(-, \mathbb{Z}_\ell)$ as well as maps of the form

$$(B.2.1) \quad \Gamma_*: H^i(X, \mathbb{Z}_\ell) \rightarrow H^{i+e}(Y, \mathbb{Z}_\ell)$$

for any $\Gamma \in H^{2d_X+e}(X \times Y, \mathbb{Z}_\ell)$. Also, we have $H^{2d_X}(X, \mathbb{Z}_\ell) \xrightarrow{\sim} \mathbb{Z}_\ell$ and therefore a pairing

$$\langle -, - \rangle: H^i(X, \mathbb{Z}_\ell) \times H^{2d_X-i}(X, \mathbb{Z}_\ell) \rightarrow H^{2d_X}(X, \mathbb{Z}_\ell) \xrightarrow{\sim} \mathbb{Z}_\ell,$$

which is perfect modulo torsion.

B.3. There is also a less well-known pairing, for any positive integer n , of the form

$$(B.3.1) \quad \langle\langle -, - \rangle\rangle_n: H^i(X, \mathbb{Z}_\ell)[\ell^n] \times H^{2d_X-i+1}(X, \mathbb{Z}_\ell)[\ell^n] \rightarrow \mathbb{Z}/\ell^n.$$

It is given by $\langle\langle \alpha, \beta \rangle\rangle_n := \langle \bar{\alpha}, \tilde{\beta} \rangle_n$, where $\bar{\alpha}$ is the mod- ℓ^n reduction of α and $\tilde{\beta}$ is any lift of β along the boundary map $\partial: H^{2d_X-i}(X, \mathbb{Z}/\ell^n) \rightarrow H^{2d_X-i+1}(X, \mathbb{Z}_\ell)$. This pairing is easily checked to be well-defined. It is not in general perfect, but its left and right kernels are

$$\begin{aligned} P_{\ell^n}^i(X) &:= H^i(X, \mathbb{Z}_\ell)[\ell^n] \cap \ell^n H^i(X, \mathbb{Z}_\ell) & \text{and} \\ P_{\ell^n}^{2d_X-i+1}(X) &:= H^{2d_X-i+1}(X, \mathbb{Z}_\ell)[\ell^n] \cap \ell^n H^{2d_X-i+1}(X, \mathbb{Z}_\ell), \end{aligned}$$

respectively, which vanish for n sufficiently large. That is, $\langle\langle -, - \rangle\rangle_n$ descends to a perfect pairing $Q_{\ell^n}^i(X) \times Q_{\ell^n}^{2d_X-i+1}(X) \rightarrow \mathbb{Z}/\ell^n$, where $Q_{\ell^n}^i(X) := H^i(X, \mathbb{Z}_\ell)[\ell^n]/P_{\ell^n}^i(X)$. (In other words, $Q_{\ell^n}^i(X)$ is the image of $H^i(X, \mathbb{Z}_\ell)[\ell^n]$ in $H^i(X, \mathbb{Z}/\ell^n)$.) One usually takes the union over all n and considers the induced perfect pairing $H^i(X)_{\text{tors}} \times H^{2d_X-i+1}(X)_{\text{tors}} \rightarrow \mathbb{Q}/\mathbb{Z}$, but we use the above formulation so that the identity (B.6.3) below makes sense.

Items (B.1)–(B.3) have the following applications for Chow motives.

B.4. Fix a positive integer a not divisible by ℓ , and let $M := (X, \varepsilon, s)$ be a Chow motive with $\mathbb{Z}[1/a]$ -coefficients; recall that this means $\varepsilon \in \text{CH}^{d_X}(X \times X)_{\mathbb{Z}[1/a]}$ is an idempotent correspondence and $s \in \mathbb{Z}$. Let $\Lambda \in \{\mathbb{Z}/\ell, \mathbb{Z}/\ell^2, \dots, \mathbb{Z}_\ell\}$. We define

$$H^i(M, \Lambda) := \text{cl}_\Lambda(\varepsilon)_* H^{i-2a}(X, \Lambda),$$

where cl_Λ denotes the cycle-class map $\text{CH}^i(-)_{\mathbb{Z}[1/a]} \rightarrow H^{2i}(-, \Lambda)$. If $N := (Y, \eta, t)$ is another Chow motive and $\varphi \in \text{Hom}(M, N)$, i.e. $\varphi \in \eta \circ \text{CH}^{2d_X+(t-s)}(X \times Y)_{\mathbb{Z}[1/a]} \circ \varepsilon$, then $\text{cl}_\Lambda(\varphi)_*$ defines a map $H^i(M, \Lambda) \rightarrow H^i(N, \Lambda)$. This defines a functor $H^i(-, \Lambda)$ on the category of Chow motives. One checks that not only is $H^i(M, \Lambda)$ functorial in the ring Λ , but also that a short exact sequence $0 \rightarrow \Lambda_1 \rightarrow \Lambda_2 \rightarrow \Lambda_3 \rightarrow 0$ induces a long exact sequence

$$\dots \rightarrow H^i(M, \Lambda_1) \rightarrow H^i(M, \Lambda_2) \rightarrow H^i(M, \Lambda_3) \rightarrow H^{i+1}(M, \Lambda_1) \rightarrow \dots$$

which is functorial in M and the short exact sequence.

B.5. Let $(-)^t$ denote the map $H^*(X \times Y, \mathbb{Z}/\ell^n) \rightarrow H^*(Y \times X, \mathbb{Z}/\ell^n)$ which is pullback along the “swap” isomorphism $Y \times X \rightarrow X \times Y$ (equivalently, pushforward along its inverse). Then

$$\langle \Gamma_* \alpha, \beta \rangle_n = (-1)^{e(i+e)} \cdot \langle \alpha, \Gamma_*^t \beta \rangle_n$$

for each $\Gamma \in H^{2d_X+e}(X \times Y, \mathbb{Z}/\ell^n)$, $\alpha \in H^i(X, \mathbb{Z}/\ell^n)$, and $\beta \in H^{2d_Y-(i+e)}(Y, \mathbb{Z}/\ell^n)$. One deduces that the same formulas hold for the pairings $\langle -, - \rangle$ and $\langle\langle -, - \rangle\rangle_n$. In particular, if $M := (X, \varepsilon, 0)$ is a Chow motive and $M^t := (X, \varepsilon^t, 0)$, then there are induced pairings

$$\begin{aligned} H^i(M, \mathbb{Z}/\ell^n) &\times H^{2d_X-i}(M^t, \mathbb{Z}/\ell^n) &\rightarrow \mathbb{Z}/\ell^n, \\ H^i(M, \mathbb{Z}_\ell) &\times H^{2d_X-i}(M^t, \mathbb{Z}_\ell) &\rightarrow \mathbb{Z}_\ell, \\ H^i(M, \mathbb{Z}_\ell)[\ell^n] &\times H^{2d_X-i+1}(M^t, \mathbb{Z}_\ell)[\ell^n] &\rightarrow \mathbb{Z}/\ell^n; \end{aligned}$$

the first is perfect, the second is perfect modulo torsion, and the third has left- and right-kernels as described in (B.3).

Finally, we describe a version of the Künneth theorem for $\mathbb{Z}_\ell$ -cohomology which gives an explicit description of the Tor terms. First, we need the following property of the maps Γ_* .

B.6. The identity $\mathrm{tr}_{X \times Y} = \mathrm{tr}_X \cdot \mathrm{tr}_Y$ implies that

$$(B.6.1) \quad (\alpha_0 \boxtimes \beta_0)_* \alpha = \langle \alpha_0, \alpha \rangle_n \cdot \beta_0$$

for any $\alpha_0 \in H^i(X, \mathbb{Z}/\ell^n)$, $\beta_0 \in H^{2d_X-i+e}(Y, \mathbb{Z}/\ell^n)$, and $\alpha \in H^j(X, \mathbb{Z}/\ell^n)$, where we put $\langle \alpha_0, \alpha \rangle_n := 0$ if α_0 and α do not live in complementary dimensions. By (B.6.1), we have

$$(B.6.2) \quad (\alpha_0 \boxtimes \beta_0)_* \alpha = \langle \alpha_0, \alpha \rangle \cdot \beta_0$$

for any $\alpha_0 \in H^i(X, \mathbb{Z}_\ell)$, $\beta_0 \in H^{2d_X-i}(Y, \mathbb{Z}_\ell)$, and $\alpha \in H^j(X, \mathbb{Z}_\ell)$. There is also an analogue of the identity (B.6.2) for the pairings $\langle\langle -, - \rangle\rangle_n$ (which can be deduced from (B.6.1)). Let $\alpha_0 \in H^i(X, \mathbb{Z}_\ell)[\ell^n]$ and $\beta_0 \in H^{2d_X-i+1}(Y, \mathbb{Z}_\ell)[\ell^n]$, and again let $\tilde{\alpha}_0$ and $\tilde{\beta}_0$ be lifts along the appropriate boundary maps. Then, for any $\alpha \in H^j(X, \mathbb{Z}_\ell)[\ell^n]$, we have

$$(B.6.3) \quad [(\partial(\tilde{\alpha}_0 \boxtimes \tilde{\beta}_0))_* \alpha] = \langle\langle \alpha_0, \alpha \rangle\rangle_n \cdot [\beta_0],$$

where ∂ is the boundary map and the brackets denote the image in the group $Q_{\ell^n}^{j+e}(Y)$ defined in (B.3).

B.7. Proposition. *Let X and Y be connected smooth projective varieties over k , and let M and N be summands of $\mathfrak{h}(X)_{\mathbb{Z}[1/a]}$ and $\mathfrak{h}(Y)_{\mathbb{Z}[1/a]}$, respectively. Fix $e \in \mathbb{Z}$. Then*

$$0 \rightarrow \bigoplus_{a+b=d_X+e} H^a(M^t, \mathbb{Z}_\ell) \otimes H^b(N, \mathbb{Z}_\ell) \xrightarrow{\boxtimes} H^{2d_X+e}(M^t \times N, \mathbb{Z}_\ell)$$

$$\xrightarrow{(-)_*} \bigoplus_j \mathrm{Hom}(H^i(M, \mathbb{Z}_\ell)_{\mathrm{tors}}, H^{i+e}(N, \mathbb{Z}_\ell)_{\mathrm{tors}}) \rightarrow 0$$

is exact, where $(-)_*$ denotes the assignment $\Gamma \mapsto \Gamma_*$ of (B.2.1).

Proof. Because this sequence is compatible with the action of correspondences, we may and do assume that $M = \mathfrak{h}(X)$ and $N = \mathfrak{h}(Y)$. The exactness in this case is a consequence of the following four facts:

- There is an exact sequence when we replace the final term by

$$\bigoplus_{a+b=d_X+e+1} \mathrm{Tor}_1^{\mathbb{Z}}(H^a(X, \mathbb{Z}_\ell)_{\mathrm{tors}}, H^b(Y, \mathbb{Z}_\ell)_{\mathrm{tors}});$$

this follows from the usual Künneth theorem in ℓ -adic cohomology.

- The groups

$$\bigoplus_i \mathrm{Hom}(H^i(X, \mathbb{Z}_\ell)_{\mathrm{tors}}, H^{i+e}(Y, \mathbb{Z}_\ell)_{\mathrm{tors}}) \quad \text{and} \quad \bigoplus_{a+b=d_X+e+1} \mathrm{Tor}_1^{\mathbb{Z}}(H^a(X, \mathbb{Z}_\ell)_{\mathrm{tors}}, H^b(Y, \mathbb{Z}_\ell)_{\mathrm{tors}})$$

have the same cardinality; this follows from the fact that $\langle\langle -, - \rangle\rangle_n$ is perfect for n sufficiently large.

- The sequence in question is a complex, i.e. $(\alpha_0 \boxtimes \beta_0)_* \alpha = 0$ for any torsion α ; this follows from the identity (B.6.2).
- The map $(-)_*$ is surjective.

The last point is a bit more complicated to explain. Fix $i \in \mathbb{Z}$, and set $A := H^i(X, \mathbb{Z}_\ell)_{\mathrm{tors}}$ and $B := H^{i+e}(Y, \mathbb{Z}_\ell)_{\mathrm{tors}}$ as well as $A_{\ell^n} := Q_{\ell^n}^i(X)$ and $B_{\ell^n} := Q_{\ell^n}^{i+e}(Y)$ in the notation of (B.3). Let H denote the image of $(-)_*$ in $\mathrm{Hom}(A, B)$. Because $\langle\langle -, - \rangle\rangle_n: A_{\ell^n} \times B_{\ell^n} \rightarrow \mathbb{Z}/\ell^n$ is perfect, the formula (B.6.3) implies that the image of H in $\mathrm{Hom}(A_{\ell^n}, B_{\ell^n})$ includes the image of the canonical map $\mathrm{Hom}(A_{\ell^n}, \mathbb{Z}/\ell^n) \otimes B_{\ell^n} \rightarrow \mathrm{Hom}(A_{\ell^n}, B_{\ell^n})$. So, the surjectivity of $(-)_*$ is a special case of the following lemma. $\blacksquare$

B.8. Lemma. *Let A and B be finite Abelian groups, and for each positive integer n , define*

$$A_n := \frac{A[n]}{A[n] \cap nA} \quad \text{and} \quad B_n := \frac{B[n]}{B[n] \cap nB}.$$

Let H be a subgroup of $\mathrm{Hom}(A, B)$ such that, for each integer n , the image of H in $\mathrm{Hom}(A_n, B_n)$ includes the image of the canonical map

$$\Phi_n: \mathrm{Hom}(A_n, \mathbb{Z}/n) \otimes B_n \rightarrow \mathrm{Hom}(A_n, B_n).$$

Then $H = \mathrm{Hom}(A, B)$.

Proof. Writing A and B as a direct sum of groups of prime-power order, we may and do assume that A and B are of ℓ -power order for some fixed prime ℓ . Fix decompositions $A = \bigoplus_{i \geq 1} A^i$ and $B = \bigoplus_{i \geq 1} B^i$ where A^i and B^i are free over $\mathbb{Z}/\ell^i$. Let us also write $A^{\leq N} := \bigoplus_{i=1}^N A^i$, and likewise $B^{\leq N}$. We will prove by induction on N that the map (defined by the fixed decompositions)

$$R_N: H \rightarrow \mathrm{Hom}(A^{\leq N}, B^{\leq N})$$

is surjective for all $N \geq 0$. The base case $N = 0$ is trivial, so assume that R_{N-1} is surjective for some $N \geq 1$. Fix $\varphi: A^{\leq N} \rightarrow B^{\leq N}$; we aim to show that φ lifts to an element of H . By the inductive hypothesis, we may and do assume that the map $A^{\leq N-1} \rightarrow B^{\leq N-1}$ induced by φ is zero. Now consider the induced decompositions

$$A_{\ell^N} = A^{\leq N-1} \oplus A^N \oplus \bigoplus_{i=1}^{N-1} \frac{\ell^i A^{N+i}}{\ell^N} =: A(1) \oplus A(2) \oplus A(3)$$

and similarly $B_{\ell^N} = B(1) \oplus B(2) \oplus B(3)$. Then Φ_{ℓ^N} factors as a direct sum of the canonical maps $\Phi_{i,j}: \text{Hom}(A(i), \mathbb{Z}/\ell^N) \otimes B(j) \rightarrow \text{Hom}(A(i), B(j))$ for $i, j \in \{1, 2, 3\}$. In particular, since A^N and B^N are free over $\mathbb{Z}/\ell^N$, $\Phi_{i,j}$ is surjective if either $i = 2$ or $j = 2$. This implies, since we already assumed that φ vanishes in $\text{Hom}(A(1), B(1))$, that there is an element in the image of Φ_{ℓ^N} whose image in $\text{Hom}(A^{\leq N}, B^{\leq N}) = \bigoplus_{i,j=1}^2 \text{Hom}(A(i), B(j))$ is φ . Since elements in the image of Φ_{ℓ^N} can be lifted to H , we are done. $\blacksquare$

REFERENCES

- [And04] Yves André, *Une Introduction aux Motifs (Motifs Purs, Motifs Mixtes, Périodes)*, Panoramas et Synthèses [Panoramas and Syntheses], vol. 17, Société Mathématique de France, Paris, 2004. MR 2115000
- [AS24] Asher Auel and V. Suresh, *Failure of the local-global principle for isotropy of quadratic forms over function fields*, Algebra Number Theory **18** (2024), no. 8, 1497–1513. MR 4856601
- [Bea09] Arnaud Beauville, *On the Brauer group of Enriques surfaces*, Math. Res. Lett. **16** (2009), no. 6, 927–934. MR 2576681
- [BGA24] Simon Brandhorst and Víctor González-Alonso, *527 elliptic fibrations on an Enriques surface*, arXiv preprint [2408.00306v1](https://arxiv.org/abs/2408.00306v1) (2024).
- [BKL76] S. Bloch, A. Kas, and D. Lieberman, *Zero cycles on surfaces with $p_g = 0$* , Compositio Math. **33** (1976), no. 2, 135–145. MR 435073
- [Blo75] Spencer Bloch, *K_2 of Artinian Q -algebras, with application to algebraic cycles*, Comm. Algebra **3** (1975), 405–428. MR 371891
- [BO74] Spencer Bloch and Arthur Ogus, *Gersten’s conjecture and the homology of schemes*, Ann. Sci. École Norm. Sup. (4) **7** (1974), 181–201. MR 412191
- [BO20] Olivier Benoist and John Christian Ottem, *Failure of the integral Hodge conjecture for threefolds of Kodaira dimension zero*, Comment. Math. Helv. **95** (2020), no. 1, 27–35. MR 4082892
- [Bon10] M. V. Bondarko, *Weight structures vs. t -structures; weight filtrations, spectral sequences, and complexes (for motives and in general)*, J. K-Theory **6** (2010), no. 3, 387–504. MR 2746283
- [CD19] Denis-Charles Cisinski and Frédéric Déglise, *Triangulated categories of mixed motives*, Springer Monographs in Mathematics, Springer, Cham, [2019] ©2019. MR 3971240
- [CDL25] François Cossec, Igor Dolgachev, and Christian Liedtke, *Enriques surfaces. I*, second ed., Springer, Singapore, 2025, with an appendix by S. Kondō. MR 4911529
- [CT95] J.-L. Colliot-Thélène, *Birational invariants, purity and the Gersten conjecture*, K-theory and algebraic geometry: connections with quadratic forms and division algebras (Santa Barbara, CA, 1992), Proc. Sympos. Pure Math., vol. 58, Part 1, Amer. Math. Soc., Providence, RI, 1995, pp. 1–64. MR 1327280
- [CT02] Jean-Louis Colliot-Thélène, *Exposant et indice d’algèbres simples centrales non ramifiées*, Enseign. Math. (2) **48** (2002), no. 1-2, 127–146, With an appendix by Ofer Gabber. MR 1923420
- [CT19] J.-L. Colliot-Thélène, *Cohomologie non ramifiée dans le produit avec une courbe elliptique*, Manuscripta Math. **160** (2019), no. 3-4, 561–565. MR 4016652
- [CTV12] Jean-Louis Colliot-Thélène and Claire Voisin, *Cohomologie non ramifiée et conjecture de Hodge entière*, Duke Math. J. **161** (2012), no. 5, 735–801. MR 2904092
- [Dia23] Humberto A. Diaz, *On the integral Hodge conjecture for varieties with trivial Chow group*, Math. Proc. Cambridge Philos. Soc. **175** (2023), no. 2, 433–443. MR 4623523
- [Fu15] Lei Fu, *Etale Cohomology Theory*, revised ed., Nankai Tracts in Mathematics, vol. 14, World Scientific Publishing Co. Pte. Ltd., Hackensack, NJ, 2015. MR 3380806

- [GG12] Sergey Gorchinskiy and Vladimir Guletskiĭ, *Motives and representability of algebraic cycles on threefolds over a field*, J. Algebraic Geom. **21** (2012), no. 2, 347–373. MR 2877438
- [GO13] Sergey Gorchinskiy and Dmitri Orlov, *Geometric phantom categories*, Publ. Math. Inst. Hautes Études Sci. **117** (2013), 329–349. MR 3090263
- [GS12] Alice Garbagnati and Matthias Schütt, *Enriques surfaces: Brauer groups and Kummer structures*, Michigan Math. J. **61** (2012), no. 2, 297–330. MR 2944482
- [Gul25] Vladimir Guletskiĭ, *The Bloch conjecture*, arXiv preprint [2512.12451v1](https://arxiv.org/abs/2512.12451v1) (2025).
- [HLOY03] Shinobu Hosono, Bong H. Lian, Keiji Oguiso, and Shing-Tung Yau, *Kummer structures on a $K3$ surface: an old question of T. Shioda*, Duke Math. J. **120** (2003), no. 3, 635–647. MR 2030099
- [HW19] Christian Haesemeyer and Charles A. Weibel, *The norm residue theorem in motivic cohomology*, Annals of Mathematics Studies, vol. 200, Princeton University Press, Princeton, NJ, 2019. MR 3931681
- [IM79] H. Inose and M. Mizukami, *Rational equivalence of 0-cycles on some surfaces of general type with $p_g = 0$* , Math. Ann. **244** (1979), no. 3, 205–217. MR 553252
- [Jan94] Uwe Jannsen, *Motivic sheaves and filtrations on Chow groups*, Motives (Seattle, WA, 1991), Proc. Sympos. Pure Math., vol. 55, Part 1, Amer. Math. Soc., Providence, RI, 1994, pp. 245–302. MR 1265533
- [Kah12] Bruno Kahn, *Classes de cycles motiviques étales*, Algebra Number Theory **6** (2012), no. 7, 1369–1407. MR 3007153
- [Kah17] ———, *Torsion order of smooth projective surfaces*, Comment. Math. Helv. **92** (2017), no. 4, 839–857, With an appendix by J.-L. Colliot-Thélène. MR 3718489
- [KS16] Bruno Kahn and Ramdorai Sujatha, *Birational motives I: Pure birational motives*, Ann. K-Theory **1** (2016), no. 4, 379–440. MR 3536433
- [lmf26] The LMFDB Collaboration, *The L-functions and modular forms database*, <https://www.lmfdb.org>, 2026.
- [Mil80] James S. Milne, *Étale Cohomology*, Princeton Mathematical Series, vol. No. 33, Princeton University Press, Princeton, NJ, 1980. MR 559531
- [Mil82] J. S. Milne, *Zero cycles on algebraic varieties in nonzero characteristic: Rojtman’s theorem*, Compositio Math. **47** (1982), no. 3, 271–287. MR 681610
- [MS82] A. S. Merkur’ev and A. A. Suslin, *K-cohomology of Severi-Brauer varieties and the norm residue homomorphism*, Dokl. Akad. Nauk SSSR **264** (1982), no. 3, 555–559. MR 659762
- [Mum65] David Mumford, *Geometric invariant theory*, Ergebnisse der Mathematik und ihrer Grenzgebiete, (N.F.), vol. Band 34, Springer-Verlag, Berlin-New York, 1965. MR 214602
- [MVW06] Carlo Mazza, Vladimir Voevodsky, and Charles Weibel, *Lecture Notes on Motivic Cohomology*, Clay Mathematics Monographs, vol. 2, American Mathematical Society, Providence, RI; Clay Mathematics Institute, Cambridge, MA, 2006. MR 2242284
- [Orl02] D. O. Orlov, *Derived categories of coherent sheaves on abelian varieties and equivalences between them*, Izv. Ross. Akad. Nauk Ser. Mat. **66** (2002), no. 3, 131–158. MR 1921811
- [PW16] C. Pedrini and C. Weibel, *Some surfaces of general type for which Bloch’s conjecture holds*, Recent advances in Hodge theory, London Math. Soc. Lecture Note Ser., vol. 427, Cambridge Univ. Press, Cambridge, 2016, pp. 308–329. MR 3409880
- [Roi80] A. A. Roitman, *The torsion of the group of 0-cycles modulo rational equivalence*, Ann. of Math. (2) **111** (1980), no. 3, 553–569. MR 577137
- [Sch98] Chad Schoen, *An integral analog of the Tate conjecture for one-dimensional cycles on varieties over finite fields*, Math. Ann. **311** (1998), no. 3, 493–500. MR 1637931
- [Sch23] Stefan Schreieder, *Refined unramified cohomology of schemes*, Compos. Math. **159** (2023), no. 7, 1466–1530. MR 4603634

- [Sch25] ———, *A moving lemma for cohomology with support*, Épijournal Géom. Algébrique ([2023–2025]), Art. 20, 50. MR 4925640
- [Ser97] Jean-Pierre Serre, *Galois Cohomology*, Springer Monographs in Mathematics, Springer-Verlag, Berlin, 1997.
- [sga77] *Cohomologie ℓ -adique et fonctions L* , Lecture Notes in Mathematics, vol. Vol. 589, Springer-Verlag, Berlin-New York, 1977, Séminaire de Géométrie Algébrique du Bois-Marie 1965–1966 (SGA 5), Edité par Luc Illusie. MR 491704
- [Sil94] Joseph H. Silverman, *Advanced Topics in the Arithmetic of Elliptic Curves*, Graduate Texts in Mathematics, vol. 151, Springer-Verlag, New York, 1994. MR 1312368
- [Sil09] ———, *The Arithmetic of Elliptic Curves*, second ed., Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR 2514094
- [sta26] The Stacks project authors, *The Stacks project*, <https://stacks.math.columbia.edu>, 2026.
- [SV23] A. N. Skorobogatov and D. Valloni, *Enriques involutions and Brauer classes*, Nagoya Math. J. **251** (2023), 606–621. MR 4637143
- [SY25] Kanetomo Sato and Takao Yamazaki, *Torsion birational motives of surfaces and unramified cohomology*, J. Inst. Math. Jussieu **24** (2025), no. 6, 2283–2315. MR 4976574
- [SZ12] Alexei N. Skorobogatov and Yuri G. Zarhin, *The Brauer group of Kummer surfaces and torsion of elliptic curves*, J. Reine Angew. Math. **666** (2012), 115–140. MR 2920883
- [Vis23a] Alexander Vishik, *On torsion spaces*, arXiv preprint [2308.14318v1](https://arxiv.org/abs/2308.14318v1) (2023).
- [Vis23b] ———, *Torsion motives*, Int. Math. Res. Not. (2023), no. 23, 20252–20295. MR 4675071
- [Voe02] Vladimir Voevodsky, *Motivic cohomology groups are isomorphic to higher Chow groups in any characteristic*, Int. Math. Res. Not. (2002), no. 7, 351–355. MR 1883180

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